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Key Points:

- We develop a linear sensitivity framework that analytically evaluates the response of wet-bulb globe temperature (WBGT) to variations in its meteorological inputs
- The framework links WBGT with dry- and wet-bulb temperature, specific humidity, and moist enthalpy in a mathematically tractable way
- This will help understand the atmospheric dynamic and thermodynamic control of WBGT changes

Supporting Information:

Supporting Information may be found in the online version of this article.

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A Linear Sensitivity Framework to Understand the Drivers of the Wet-Bulb Globe Temperature Changes

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Abstract Better understanding of the physical drivers of sufficiently realistic representation of human heat stress is crucial for improving prediction and enhancing preparedness. Wet-bulb globe temperature (WBGT) is a standard metric for workplace heat stress; however, its calculation involves complex parameterizations of radiative and convective energy exchange, making it difficult to understand the driving mechanisms behind WBGT changes. To address this issue, we introduce a sensitivity framework to analytically evaluate WBGT's response to meteorological input changes. By examining the form of sensitivity coefficients, we gain insights into the interactive effects of multiple environmental parameters in controlling WBGT. Given constant wind and solar radiation, the natural wet-bulb and black globe temperatures change at the same rate and direction as the wet- and dry-bulb temperatures, despite considerable differences in their absolute values. The framework, while having state-dependent sensitivity coefficients, can be linearized, transforming WBGT into a linear combination of temperature, specific humidity, surface pressure, and terms representing wind and solar radiation effects. These explicit and mathematically tractable relations between WBGT and more intuitively understandable variables enable leveraging established theories and methods to understand the driving mechanisms of WBGT changes. We apply the framework to understand the physical drivers of regional WBGT scaling with global warming and extreme WBGT synoptic events. The sensitivity framework also provides a customizable approach to develop locally tuned linear approximations of WBGT, with clear expectations regarding the direction and magnitude of the induced biases. It can also be used to diagnose sources of biases in existing simplified WBGT approximations.

Plain Language Summary Better understanding of the physical causes of human heat stress is important for improving prediction and preparing society. The reliability of such understanding depends on a reasonable representation of heat stress. Wet-bulb globe temperature (WBGT) is a standard metric for workplace heat stress. But its calculation involves complex treatment of energy exchange between the instrument and the ambient environment, which makes it difficult to understand the driving mechanisms behind WBGT changes. To address this challenge, we introduce a framework that can establish explicit and mathematically tractable relations between WBGT and several other intuitively understandable variables. This enables us to leverage established theories and methods to understand the driving mechanisms of WBGT changes. We apply the framework to understand the physical processes that control how WBGT increases regionally with global warming and that lead to extremely high WBGT weather conditions. The framework also offers a flexible way to develop locally tailored, simplified versions of WBGT with clear expectations about the direction and magnitude of any biases introduced. It can also help identify sources of errors in existing simplified WBGT calculations.

1. Introduction

Heat stress threatens human health (Buzan & Huber, 2020; Ebi et al., 2021; Kjellstrom et al., 2016) with wide-ranging societal (Burke et al., 2018; Cane et al., 2014; Hsiang et al., 2013) and economic impact (Burke et al., 2015; Orlov et al., 2020; Saeed et al., 2022). A good understanding of the physical processes controlling heat stress is important for improving its prediction and informing heat mitigation and adaptation strategies. In order to be more practically useful, the physical understanding needs to be generated from metrics that can reasonably represent human heat stress.

Numerous studies have investigated the driving mechanism of hot temperature events including both local and remote processes across the atmosphere, land and ocean (Domeisen et al., 2022; Perkins, 2015). However, human heat stress is also influenced by humidity, wind, and radiation (Buzan & Huber, 2020; Foster et al., 2021; Parsons, 2014), which may vary independent of the temperature. Additionally, heat stress is influenced by the climate processes that modify those variables. For example, soil moisture deficits reallocate surface radiative energy from latent to sensible heating which raises air temperature but lowers humidity (Seneviratne et al., 2010).

Digging into climate processes, several studies have incorporated the effects of humidity in exploring the governing processes of heat stress such as proximity to nearby moisture sources (Monteiro & Caballero, 2019), anomalously high surface evapotranspiration (Guo et al., 2022; Mishra et al., 2020; Souri et al., 2020), a shallower atmosphere boundary layer (Kong & Huber, 2023; Raymond et al., 2021), large-scale modes of variability such as the ENSO (Y. Zhang et al., 2024) and Madden-Julian Oscillation (C. C. Ivanovich et al., 2024), and in the tropics remote control from warm SST elsewhere (Y. Zhang et al., 2021). In these studies, heat stress is typically represented by thermodynamic quantities such as wet-bulb temperature (T_w) or moist enthalpy because of their straightforward dynamic and thermodynamic constraint (Byrne & O'Gorman, 2013, 2016; Findell & Eltahir, 2003), making them amenable to theoretical understanding.

However, these thermodynamic quantities are not intended for or well calibrated to human heat stress, and miss the effects of wind and radiation (Foster et al., 2022; Havenith et al., 2024; Lu & Romps, 2023a; Vanos et al., 2023). This diminishes the practical relevance of the physical insights generated from such quantities, and may potentially misguide heat mitigation policies. For example, several studies have found irrigation to exacerbate heat stress as measured by T_w or moist enthalpy possibly due to a shallower boundary layer and less free-troposphere dry air entrainment into the boundary layer under wetter soil conditions (Guo et al., 2022; Kong & Huber, 2023; Krakauer et al., 2020; Mishra et al., 2020). However, this finding largely depends on the weighting of humidity in the adopted heat stress metrics (Chakraborty et al., 2025; Kong & Huber, 2023), and T_w or moist enthalpy has been criticized for its over-emphasis on humidity especially under hot-dry conditions (Lu & Romps, 2023a, 2023b; Simpson et al., 2023; Vanos et al., 2023). This raises questions about the robustness of heat stress amplification by irrigation when employing a less humidity-weighted metric that may better capture the effects of heat stress on the human body (Chakraborty et al., 2025; Parajuli et al., 2024; S. Sun et al., 2024).

Some examples of these potentially "better" heat stress metrics include the heat index (Lu & Romps, 2022; Rothfusz, 1990; Steadman, 1979), universal thermal climate index (UTCI) (Bröde et al., 2012) and wet-bulb globe temperature (WBGT) (Yaglou & Minard, 1957), which are supported by empirical calibrations and/or human thermal physiology principles. However, it is challenging to understand—in a quantitative sense—the physical constraint on variations in these "better" metrics due to their complex formulations, especially in the case of the heat index and UTCI with complex parameterization of human thermoregulatory response and/or heavy empirical fitting. Overcoming these challenges may be somewhat easier in the case of WBGT, as it can be explicitly derived from solving the energy balance equations of the passive sensors (Liljegren et al., 2008). WBGT incorporates the effects of all four meteorological parameters, and has been demonstrated to be as good or better than alternative metrics in predicting human heat stress compensability (Vecellio et al., 2022), assessing the physiological influences of heat stress (Ioannou et al., 2022), modeling heat-related health outcomes (e.g., heat stroke morbidity) in Japan (Guo et al., 2024), and capturing the interactive effects of multiple meteorological factors on human physical work capacity (Foster et al., 2022; Havenith et al., 2024).

The goal of this work is to advance the understanding of the physical drivers of WBGT changes. This is achieved by developing a sensitivity framework that analytically evaluates the response of WBGT to variations in temperature, specific humidity, wind, solar radiation, and surface pressure. The sensitivity coefficients, although originally state-dependent, become constants after the framework is linearized against a certain reference point. This renders WBGT as a linear combination of these physically more intuitive meteorological variables which are also commonly available from observations. With such a framework, existing theories and methods on constraining changes in these meteorological variables can be applied to interpreting WBGT changes and understanding the driving mechanisms.

The remainder of this paper is structured as follows. Section 2 introduces the sensitivity framework (Section 2.1), inspects state-dependent behavior of the sensitivity coefficients (Section 2.2), presents a simplified case with constant solar radiation and wind speed (Section 2.3), and applies linearization to the framework (Section 2.4). Section 3 presents two example applications of the framework to understanding the driving mechanism of WBGT changes. Section 4 compared the linear framework with a simplified linear approximation of WBGT that is widely used in the literature. A brief summary and discussions are made in Section 5.

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2. A Linearized Sensitivity Framework of the Wet-Bulb Globe Temperature

2.1. Introducing the Framework

WBGT is defined as

$$WBGT = 0.7T_{nw} + 0.2T_g + 0.1T_a \tag{1}$$

under outdoor conditions where T_{nw}, T_g and T_a refer to natural wet-bulb temperature, black globe temperature and dry-bulb temperature respectively (Yaglou & Minard, 1957). The "gold standard" physics-based WBGT model developed by Liljegren et al. (2008) solves T_{nw} and T_g as the equilibrium skin temperatures of the wet bulb and black globe sensors in maintenance of energy balance. In this section, we analytically derive the sensitivities of T_g, T_{nw} and consequently WBGT to variations in each meteorological parameter starting from the energy balance equations of the black globe and wet bulb.

2.1.1. Black Globe Temperature

The energy balance of the black globe is given by

$$\sigma\epsilon_g T_g^4 + h_{cg}(T_g - T_a) = LR_g + SR_g \tag{2}$$

where energy gain from long-wave (LR_g) and short-wave radiation (SR_g) reaching the globe is balanced by cooling from outgoing long-wave radiation and convection corresponding to the two terms on the left side of Equation 2. h_{cg} represents convective heat transfer coefficient between the black globe and ambient air. σ and ϵ_g denote the Stefan-Boltzmann constant and emissivity of the black globe.

SR_g and LR_g combine the effects of surface downward and upwelling short- and long-wave radiation (please refer to Liljegren et al. (2008) and Kong and Huber (2024) for their formulations). Liljegren et al. (2008)'s original model only includes surface downward solar radiation as input and approximates other radiation components from air temperature, humidity and ground albedo. Kong and Huber (2022a) modified this model by explicitly including all radiation components when available, for example, from climate model outputs or reanalysis products. For the purpose of developing the sensitivity framework, here we approximate LR_g from air and ground temperature following Liljegren et al. (2008).

$$LR_g = \frac{1}{2}\epsilon_g\sigma\left(\epsilon_a T_a^4 + \epsilon_{sfc} T_{sfc}^4\right)$$

where ϵ_a and ϵ_{sfc} represent emissivity of the air and ground surface; T_{sfc} is ground surface temperature. We assume that the effective radiating temperature of the ground surface is equal to the air temperature $\left(T_{sfc,eff}^4 = \epsilon_{sfc} T_{sfc}^4 \approx T_a^4\right)$ as in Liljegren et al. (2008), and the air is close to black body $(\epsilon_a \approx 1)$. This allows LR_g to be simplified into

$$LR_g \approx \sigma\epsilon_g T_a^4$$

which is equivalent to saying that the effects of long-wave radiative heating can be represented by a mean radiant temperature of T_a in the absence of solar radiation. This assumption may not hold as effectively over deserts where the air significantly deviates from the black body, or in workplaces with notable sources of thermal radiation such as aluminum potrooms. With this assumption, Equation 2 can be rearranged to

$$(h_{cg} + h_{rg})(T_g - T_a) = SR_g \tag{3}$$

where $h_{rg} = \sigma\epsilon_g\left(T_g^2 + T_a^2\right)(T_g + T_a)$ can be interpreted as a thermal radiative heat transfer coefficient. The physical interpretation of Equation 3 is that the efficiency of energy loss through convection (h_{cg}) and long-wave cooling (h_{rg}) modulates the required temperature gradient between the globe and ambient air in order to balance

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the energy gain from solar radiation (Kong & Huber, 2024). Taking differential on both sides of Equation 3, we have the following:

$$dT_g = \alpha_{T_a} dT_a + \alpha_{SR} dSR_g + \alpha_h (dh_{cg} + dh_{rg}) \tag{4}$$

$$\alpha_{T_a} = 1 \tag{4.1}$$

$$\alpha_{SR} = \frac{1}{h_{cg} + h_{rg}} \tag{4.2}$$

$$\alpha_h = -\frac{T_g - T_a}{h_{cg} + h_{rg}} \tag{4.3}$$

The coefficients α_{T_a}, α_{SR} and α_h evaluate the sensitivity of T_g to variations in temperature, solar radiative heating and the efficiency of heat transfer through convection or thermal radiation.

The solar radiative heating term (SR_g) combines multiple factors including downward solar radiation (both direct and diffuse components), solar zenith angle, ground albedo, and the globe geometry. However, all factors except downward solar radiation either can be considered static or follow a constant diurnal cycle. Therefore, dSR_g can be considered to represent the effects of changes in solar radiation. h_{cg}, as derived from empirical correlations under forced convection (Liljegren et al., 2008), is primarily influenced by wind speed (Kong & Huber, 2024). Therefore, dh_{cg} can be used to represent the effects of wind speed changes. h_{rg} is a function of T_g and T_a. However, it only varies by 0.5% per °C change in T_g or T_a, and is typically considerably less than h_{cg} (convection is the dominant energy loss pathway for the black globe) (Kong & Huber, 2024). As a result, the effects of dh_{rg} is expected to be small and can be neglected, simplifying Equation 4 to

$$dT_g = \alpha_{T_a} dT_a + \alpha_{SR} dSR_g + \alpha_h dh_{cg} \tag{5}$$

which assesses the responses of T_g to changes in temperature, solar radiation, and wind.

2.1.2. Natural Wet-Bulb Temperature

The energy balance of the wet bulb is represented by

$$\sigma \epsilon_w T_{nw}^4 + h_{cw}(T_{nw} - T_a) + k_x \frac{e_w - e_a}{P_s - e_w} M_{H2O} L_v = LR_w + SR_w \tag{6}$$

where energy gain from long-wave (LR_w) and short-wave (SR_w) radiation reaching the wick is balanced by energy loss through outgoing long-wave radiation, convection and evaporation corresponding to three terms on the left side of Equation 6. k_x denotes mass transfer coefficient which is linked to convective heat transfer coefficient (h_{cw}) via the Chilton-Colburn analogy (Chilton & Colburn, 1934), and both parameters are predominantly affected by wind speed (Kong & Huber, 2024). ϵ_w, M_{H2O} and L_v denote emissivity of the wick, the molecular weight of water vapor, and the heat of vapourization. e_a and e_w represent vapor pressure of the ambient air and a saturated air at the temperature of T_{nw}.

SR_w and LR_w combine the effects of surface downward and upwelling short- and long-wave radiation (please refer to Liljegren et al. (2008) and Kong and Huber (2024) for their formulations). Similar to the case of T_g, LR_w is approximated as $LR_w \approx \sigma \epsilon_w T_a^4$ for the purpose of developing the sensitivity framework. In addition, vapor pressure can be approximated by $e_a = P_s r/(r + \zeta) \approx P_s q/\zeta$, where P_s, r and q denote surface pressure, mixing ratio and specific humidity, respectively, and $\zeta = 0.622$ is the ratio between the gas constants for dry air and water vapor. We also assume $P_s - e_w \approx P_s$. These approximations allow Equation 6 to be rewritten as:

$$(h_{cw} + h_{rw})(T_{nw} - T_a) + \kappa(q_w - q) = SR_w \tag{7}$$

where $h_{rw} = \sigma\epsilon_w(T_{nw}^2 + T_a^2)(T_{nw} + T_a)$ is a thermal radiative heat transfer coefficient, $\kappa = k_x M_{H2O} L_v/\zeta$ is a scaled mass transfer coefficient, and q_w is saturated specific humidity at the temperature of T_{nw}. Taking differential on both sides of Equation 7 and noting that

$$dq_w = (\partial q_w/\partial P_s)\,dP_s + (\partial q_w/\partial T_{nw})\,dT_{nw} \tag{8}$$

we have

$$dT_{nw} = \beta_{T_a} dT_a + \beta_q dq + \beta_{SR} dSR_w + \beta_\kappa d\kappa + \beta_h (dh_{cw} + dh_{rw}) + \beta_{P_s} dP_s \tag{9}$$

$$\beta_{T_a} = (h_{cw} + h_{rw})/\xi \tag{9.1}$$

$$\beta_q = \kappa/\xi \tag{9.2}$$

$$\beta_{SR} = 1/\xi \tag{9.3}$$

$$\beta_\kappa = (q - q_w)/\xi \tag{9.4}$$

$$\beta_h = (T_a - T_{nw})/\xi \tag{9.5}$$

$$\beta_{P_s} = -\kappa(\partial q_w/\partial P_s)/\xi \tag{9.6}$$

$$\xi = h_{cw} + h_{rw} + h_{ew} \tag{9.7}$$

$$h_{ew} = \kappa(\partial q_w/\partial T_{nw}) \tag{9.8}$$

where h_{ew} represents the sensitivity of evaporative energy loss to changes in T_{nw}, or in other words, the strength of negative feedback from increased evaporative cooling as T_{nw} rises. Another way to interpret h_{ew} is to treat it as a coefficient that measures the efficiency of evaporative heat transfer across a temperature gradient between the wick temperature and the ambient dewpoint temperature (T_d)

$$\kappa(q_w - q) \approx h_{ew}(T_{nw} - T_d)$$

As demonstrated by Kong and Huber (2024), h_{ew} is usually much larger than both h_{cw} and h_{rw}. In combination with the larger temperature gradient it works on ($T_{nw} - T_d \geq T_{nw} - T_a$), evaporation is clearly the dominant energy loss pathway for the wet bulb.

Within Equation 9, the effect of dh_{rw} is typically small (as in the case of T_g) and can be neglected:

$$dT_{nw} = \beta_{T_a} dT_a + \beta_q dq + \beta_{SR} dSR_w + (\beta_\kappa d\kappa + \beta_h dh_{cw}) + \beta_{P_s} dP_s \tag{10}$$

where the combination of $d\kappa$ and dh_{cw} represent the effects of wind speed changes. With that, Equation 10 evaluates the response of T_{nw} to changes in temperature, specific humidity, solar radiation, wind, and surface pressure.

2.1.3. Wet-Bulb Globe Temperature

Combining Equations 1, 5, and 10, we derive the response of WBGT to changes in temperature, specific humidity, solar radiation, wind and surface pressure corresponding to the five terms on the right side of Equation 11.

$$\begin{aligned} dWBGT = {} & \overbrace{\gamma_{T_a} dT_a}^{\text{Temperature}} + \overbrace{\gamma_q dq}^{\text{humidity}} + \overbrace{(0.7\beta_{SR} dSR_w + 0.2\alpha_{SR} dSR_g)}^{\text{Solar radiation}} \\ & + \underbrace{(0.7\beta_\kappa d\kappa + 0.7\beta_h dh_{cw} + 0.2\alpha_h dh_{cg})}_{\text{Wind}} + \underbrace{0.7\beta_{P_s} dP_s}_{\text{Surface pressure}} \end{aligned} \tag{11}$$

where $\gamma_{T_a} = 0.3 + 0.7\beta_{T_a}$ and $\gamma_q = 0.7\beta_q$.

The sensitivity coefficients in Equation 11 measure the response of WBGT to variations in each variable while holding other variables constant. In reality, variables on the right side of Equation 11 are not physically independent and tend to covary. For instance, changes in solar radiation affect the ground surface energy balance, altering surface sensible and latent heat fluxes into the atmosphere, and therefore near-surface air temperature and humidity. However, these effects are embedded within the dT_a and dq terms. By holding T_a and q constant, the effect of solar radiation changes in Equation 11 only considers the direct radiative heating from the radiation beam reaching the sensors. Similarly, the impact of surface pressure changes on the specific humidity of ambient air is included within the dq term. The dP_s term in Equation 11 only accounts for its impact on saturated specific humidity at the wick surface (Equation 8).

2.2. Sensitivity Coefficients

The sensitivity coefficients derived above warrant particular inspection as they evaluate the response of WBGT to variations in each meteorological input, and demonstrate whether and how this response depends on the values of other variables. In this section, we explore the state-dependent behavior of the coefficients across a multi-dimensional climatic phase space (Figure 1). Note that Liljegren et al. (2008)'s original formulation is employed here and similarly for Figures 2–4. Namely, only total downward solar radiation is included as input, with other radiation components being approximated. This approach is taken because thermal radiation, and direct and surface-reflected solar radiation inherently depend on temperature and total downward solar radiation, making it inappropriate to treat these other radiation components as independent dimensions of the climatic phase space. In contrast, all radiation components are taken from climate model output within example applications of the framework based on reanalysis or climate model simulations (Section 3 and 4).

2.2.1. T_g Coefficients

The effect of T_a changes (quantified by α_{T_a}) is independent of other parameters. Although keeping constant solar radiation and wind speed, T_g varies at the same rate and in the same direction as T_a (Equation 4.1). This ensures a constant temperature gradient $(T_g - T_a)$ and therefore a fixed amount of cooling through convection and thermal radiation that balances solar radiative heating. When solar radiation is absent $(SR_g = 0)$, T_g approaches T_a (Equation 3) as is typically assumed when calculating WBGT for nighttime or indoor environment (Lemke & Kjellstrom, 2012). In reality, T_g is lower than T_a at nighttime because the air is not a black body which causes net cooling from the long-wave radiative exchange between the black globe and ambient air (Kong & Huber, 2024).

A stronger solar radiation raises T_g the magnitude of which (as measured by α_{SR}) is inversely proportional to the overall heat transfer efficiency via convection and thermal radiation (Equation 4.2), and therefore inversely related to wind speed given the predominant role of convection in energy loss. Since $h_{cg} \propto V^{0.5}$ where V is wind speed (Liljegren et al., 2008), $\alpha_{SR} \propto V^{-0.5}$ suggesting that the impact of incremental increases in solar radiative heating decreases in a decelerating manner as wind speed increases (Figure 1a).

Increasing wind raises h_{cg}, which cools T_g when $T_g > T_a$ ($\alpha_h < 0$) as is typically the case in daytime. This cooling effect (as measured by α_h) is directly proportional to the temperature gradient between the black globe and ambient air, and inversely proportional to heat transfer efficiency (Equation 4.3). Therefore, cooling benefits from per unit increase in h_{cg} are most pronounced under low wind and strong solar radiation (Figure 1b). Note that per unit enhancement in wind speed is expected to cause larger increases in h_{cg} under weaker wind given $h_{cg} \propto V^{0.5}$ (Kong & Huber, 2024; Liljegren et al., 2008). This indicates strong marginal cooling benefits from incremental increases of wind speed from a low baseline value.

2.2.2. T_{nw} Coefficients

The sensitivity of T_{nw} to T_a variations (β_{T_a}) depends on the relative efficiency of heat transfer through evaporation compared with that through convection and thermal radiation (Equation 9.1). If $h_{ew} = 0$ (either $\kappa = 0$ or $\partial q_w / \partial T_{nw} = 0$), evaporative cooling is either absent or does not depend on the value of T_{nw} (negative feedback from evaporative cooling is absent). In both cases, T_{nw} needs to follow T_a increases—while keeping other variables constant—in a 1:1 fashion $(\beta_{T_a} = 1)$ to maintain a constant temperature gradient that satisfies energy

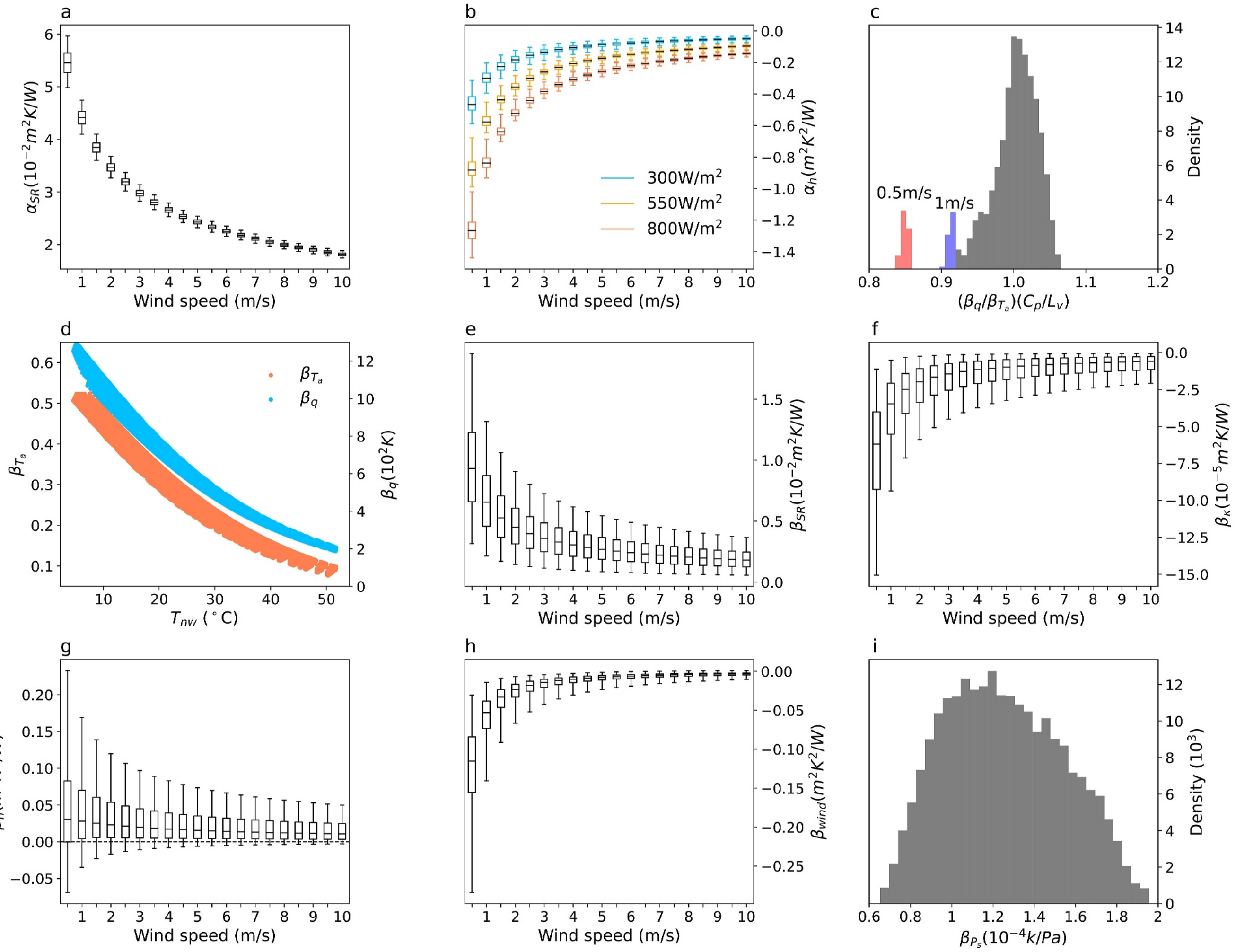

Figure 1. Variation in panels (a) α_{SR}, (b) α_h, (e) β_{SR}, (f) β_κ, (g) β_h, and (h) β_{wind} as functions of wind speed. (c) Histogram of β_q/β_{T_a} normalized by L_v/C_p; the red and blue bars correspond to wind speeds of 0.5 and 1 m/s. (d) β_{T_a} and β_q as functions of T_{nw}. (i) Histogram of β_{P_s}. All panels are constructed from a multi-dimensional climatic phase space with T_a ranging from 15 to 50°C, relative humidity from 10% to 100%, wind speed from 0.5 to 10 m/s, solar radiation from 300 to 800 W/m^2, and surface pressure from 950 to 1050 hPa. Panel (b) is grouped by solar radiation levels of 300, 550, and 800 W/m^2. The boxes within all boxplots extend from the lower to upper quartile, with a line at the median. The whiskers represent the minimum and maximum values. Both the cosine value of the solar zenith angle and the proportion of direct solar radiation are fixed at 0.75. Long-wave and surface reflected short-wave radiation are approximated from temperature, humidity and an assumed ground albedo of 0.45 as in Liljegren et al. (2008).

balance. If $h_{ew} > 0$ as in the reality, a higher T_{nw} facilitates evaporative cooling by enlarging the humidity gradient between the saturated wick and the ambient air. This negative feedback reduces the required enhancement in T_{nw} in achieving equilibrium $\left(\beta_{T_a} < 1\right)$.

β_{T_a} can be simplified making use of the Lewis relation (Lewis, 1922), $LR = \kappa/h_{cw}$, which states an analogy between mass and heat transfer (Chilton & Colburn, 1934). For the air-water system, $LR \approx L_v/C_p$ where C_p is the specific heat capacity of dry air at constant pressure (Gagge & Gonzalez, 1996). Since h_{rw} is usually considerably smaller than h_{cw} (Kong & Huber, 2024), we assume

$$\frac{\kappa}{h_{cw} + h_{rw}} \approx \frac{L_v}{C_p} \tag{12}$$

as confirmed in Figure 1c. The more evident violation of this relation in Figure 1c corresponds to relatively low wind speed (0.5 m/s) where h_{rw} is more comparable to h_{cw}. This simplifies β_{T_a} into

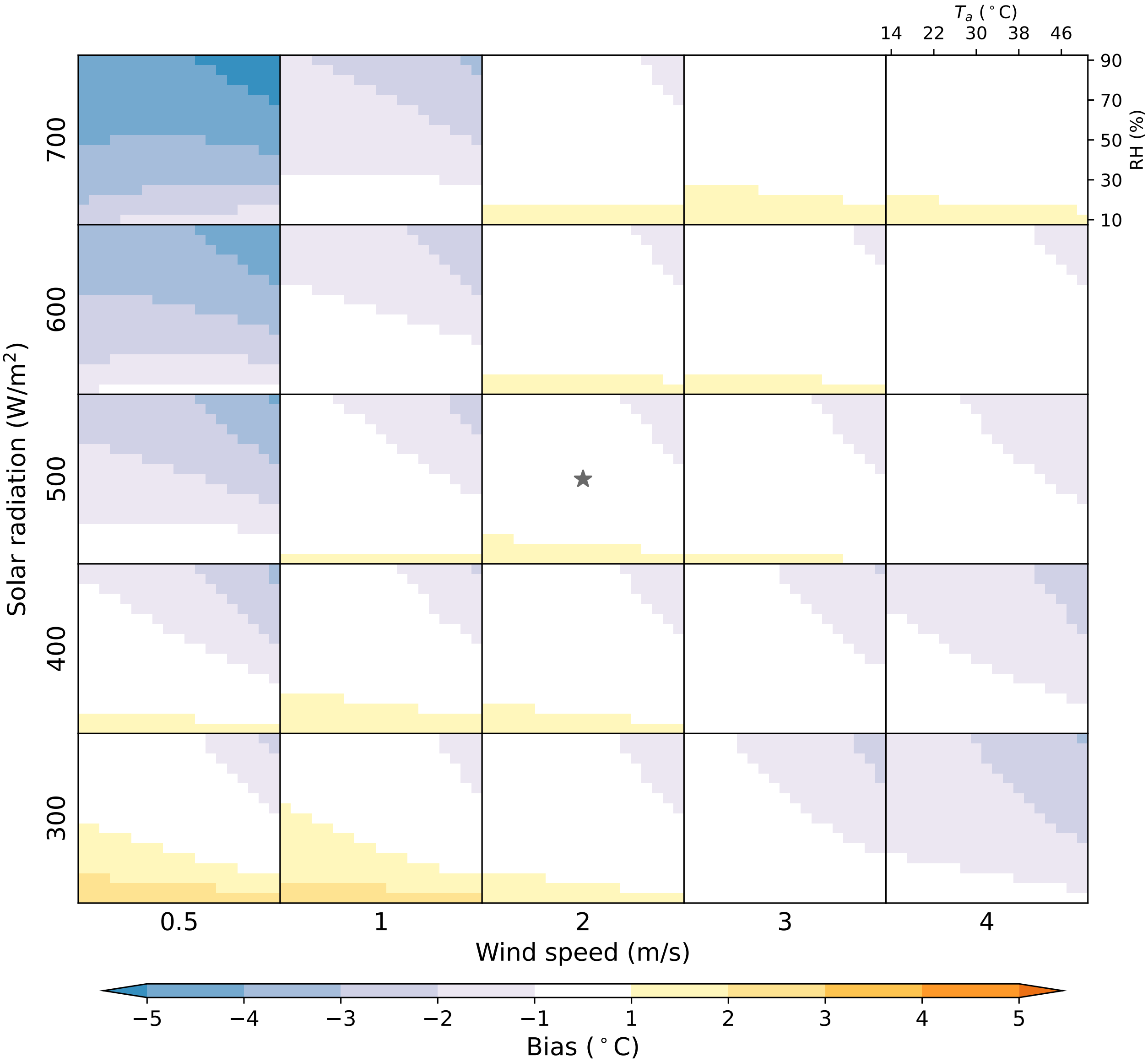

Figure 2. Biases in pseudo T_g across a range of T_a (14–50°C), relative humidity (RH) (10%–90%), wind speed (0.5, 1, 2, 3, and 4 m/s), and solar radiation (300, 400, 500, 600, and 700 W/m^2). Bias is calculated as the pseudo value minus the true value. Each small box represents a 2D T_a-RH phase space under fixed levels of solar radiation and wind speed. Sensitivity coefficients are calculated against a reference point (marked by the star) with T_a of 32°C, RH of 50%, wind speed of 2 m/s, and solar radiation of 500 W/m^2. Surface pressure is set at 1,000 hPa, and both the cosine value of the solar zenith angle and the proportion of direct solar radiation are fixed at 0.75. Long-wave and surface reflected short-wave radiation are approximated from temperature, humidity and an assumed ground albedo of 0.45 as in Liljegren et al. (2008).

$$\beta_{T_a} \approx \frac{C_p}{C_p + L_v \partial q_w / \partial T_{nw}} \tag{13}$$

Similarly, the q sensitivity coefficient $\left(\beta_q\right)$ can be rewritten as follows:

$$\beta_q \approx \frac{L_v}{C_p + L_v \partial q_w / \partial T_{nw}} \tag{14}$$

Therefore, both β_{T_a} and β_q decrease with increasing T_{nw} (Figure 1d) due to the increasingly strong negative feedback from evaporative cooling as a result of the exponential Clausius-Clapeyron scaling. Since $\beta_{T_a}/\beta_q \approx C_p/L_v$, T_{nw} has the same relative weights on T_a and q as moist enthalpy or T_w. In fact, when solar

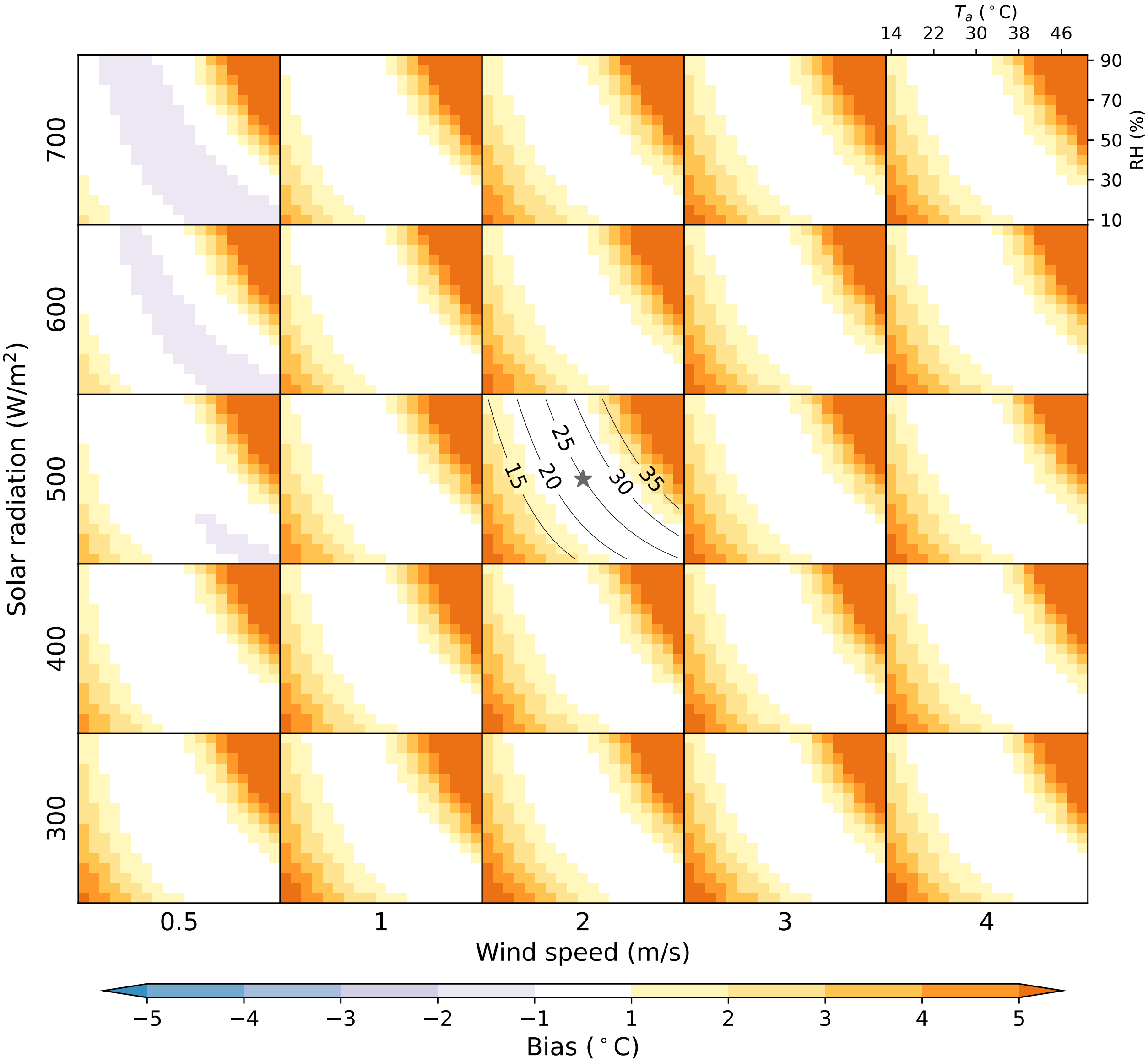

Figure 3. Same as Figure 2 but for T_{nw}. The contours denote T_{nw} isolines under reference level solar radiation and wind speed.

radiation is absent, the energy balance of the wet bulb as shown in Equation 7 becomes identical to the definition of T_w

$$C_p(T_{nw} - T_a) + L_v(q_w - q) = 0 \tag{15}$$

where Equation 12 is used. Namely, T_{nw} approaches T_w as solar radiation drops to zero.

Solar radiation raises T_{nw} above T_w the effect of which (quantified by β_{SR}) is inversely proportional to the total heat transfer efficiency (ξ), hence inversely related to wind speed (Figure 1e). Since both h_{cw} and κ are proportional to $V^{0.6}$ (Liljegren et al., 2008), $\beta_{SR} \propto V^{-0.6}$ indicating that the effect of additional solar radiative heating decreases at a progressively slower rate as wind speed increases (Figure 1e), similar to the case of T_g. Note that the total heat transfer efficiency ξ also depends on T_{nw} (Equations 9.7 and 9.8), which helps explain the larger spread in β_{SR} compared with α_{SR} when plotted against wind speed (compare Figures 1a and 1e).

The effects of wind speed changes depend on whether solar radiation is present or not. If $SR_w = 0$

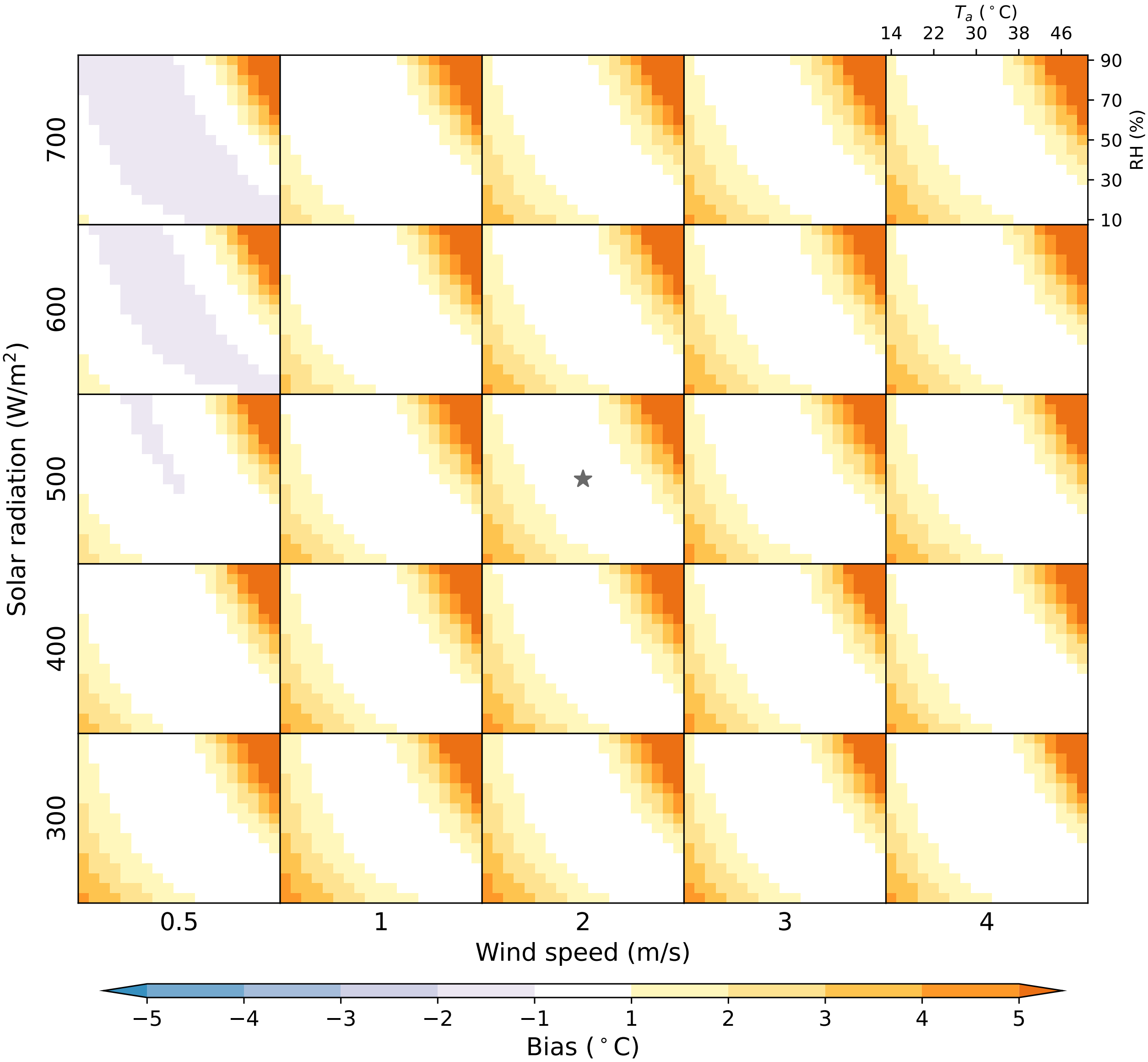

Figure 4. Same as Figure 2 but for wet-bulb globe temperature.

$$\beta_\kappa d\kappa + \beta_h dh_{cw} \approx -\frac{d\kappa}{L_v \xi}\left(C_p(T_{nw} - T_a) + L_v(q_w - q)\right) = 0$$

which can be derived by combining Equations 9.4 and 9.5 and the Lewis relation. The physical interpretation is that changes in wind speed only alter the time needed to reach equilibrium without changing the equilibrium temperature of the wet wick which, as demonstrated above, is T_w. This is associated with a balance between latent cooling and sensible heating that is invariant to wind speed variations.

After introducing solar radiation, T_{nw} rises, which enhances evaporative cooling by enlarging the wick-environment vapor pressure gradient while reducing sensible heating from convection by weakening the temperature gradient. As a result, evaporation and convective heat transfer together produce cooling to balance the solar radiative heating. In fact, the wick can lose energy from convective heat transfer if T_{nw} is heated above T_a, such as under strong solar radiation and low wind speeds (corresponding to negative β_h in Figure 1g). Employing the Lewis relation, the net effect of wind speed changes becomes

$$\beta_\kappa d\kappa + \beta_h dh_{cw} = \beta_{wind} dh_{cw}$$

$$\beta_{wind} \approx \beta_\kappa L_v / C_p + \beta_h = -\frac{1}{C_p \xi}\left[C_p (T_{nw} - T_a) + L_v (q_w - q)\right]$$

where β_{wind} evaluates the combined effects of changes in convective heat and mass transfer efficiency due to wind speed changes. Note that $\beta_{wind} < 0$ when $SR_w > 0$, suggesting a stronger cooling from convective heat and mass transfer as wind speed increases. β_{wind} has larger magnitudes under weaker wind (Figure 1h), indicating larger cooling effects from per unit increase in h_{cw} under low wind conditions. Similar to the case of T_g, per unit increase in wind speed is expected to induce larger increases in h_{cw} under weaker wind (Kong & Huber, 2024). This once again underscores stronger marginal cooling benefits from incremental wind speed increases from lower baseline values.

However, it needs to be mentioned that the parameterizations of convective mass and heat transfer in Liljegren et al. (2008)'s model are based on empirical correlations under forced convection, and is not valid under very low wind speed (e.g., <0.2 m/s) when convection is primarily driven by buoyancy (free convection). In fact, the estimation of T_{nw} under free convection is subject to the problem of multiple steady state solutions due to the strong nonlinear structure of heat and mass transfer equations (d'Ambrosio Alfano et al., 2012).

The sensitivity coefficient to surface pressure variations $\left(\beta_{P_s}\right)$ can be simplified to

$$\beta_{P_s} = -\frac{L_v \partial q_w / \partial P_s}{C_p + L_v \partial q_w / \partial T_{nw}}$$

by combining Equations 9.6 and 12. Therefore, the values of β_{P_s} mainly depend on T_{nw} and P_s, and are on the order of 10^{-4} (Figure 1i). This suggests that a 10 hPa change in P_s (while holding q constant) only varies T_{nw} by around 0.1°C. Additionally, β_{P_s} shows a relatively small range of variation from 0.6 to 2 $\times 10^{-4}$, across a wide range of environmental conditions (Figure 1i). Therefore, the effects of dP_s are expected to be small and approximately linear.

Please be aware that all the simplifications using Lewis relation are only meant to help interpret the state-dependent behavior of these sensitivity coefficients. Their original form will still be used in practical applications for the sake of accuracy.

2.3. A Simplified Case: Constant Solar Radiation and Wind Speed

In the preceding section, we demonstrated that in the absence of solar radiation, T_g and T_{nw} converge to T_a and T_w respectively. Solar radiation drives T_g and T_{nw} away from T_a and T_w. The magnitude of the deviation is primarily modulated by wind speed, as indicated by α_{SR} (Equation 4.2) and β_{SR} (Equation 9.3), although β_{SR} also depends on T_{nw} itself. This implies that, if solar radiation and wind speed remain unchanged, the deviation of T_g and T_{nw} away from T_a and T_w should remain relatively stable. Namely, despite considerable differences in absolute values, changes in T_g and T_{nw} may be close to changes in T_a and T_w.

With constant solar radiation and wind speed, Equations 5 and 10 are simplified to

$$dT_g = dT_a \tag{16}$$

$$dT_{nw} = \frac{C_p}{C_p + L_v \partial q_w / \partial T_{nw}} dT_a + \frac{L_v}{C_p + Lv \partial q_w / \partial T_{nw}} dq - \frac{L_v \partial q_w / \partial P_s}{C_p + L_v \partial q_w / \partial T_{nw}} dP_s \tag{17}$$

where the simplified forms of β_{T_a}, β_q and β_{P_s} from Lewis relation are used. T_g follows T_a changes in a 1:1 fashion as described earlier. The similarity between dT_{nw} and dT_w will become evident after dT_w is explicitly derived.

T_w is defined by

$$C_p (T_w - T) + L_v (q^* - q) = 0 \tag{18}$$

where q^* refers to saturated specific humidity at the temperature of T_w. Taking differential on both sides of the equation, we have

$$dT_w = \frac{C_p}{C_p + L_v \partial q^* / \partial T_w} dT + \frac{L_v}{C_p + Lv \partial q^* / \partial T_w} dq - \frac{L_v \partial q^* / \partial P_s}{C_p + L_v \partial q^* / \partial T_w} dP_s \quad (19)$$

The similarity between dT_{nw} and dT_w is clear by comparing Equations 17 and 19. The only difference is whether the partial derivatives are evaluated at the temperature of T_{nw} or T_w. Note that the partial derivative of saturated specific humidity with respect to either temperature or surface pressure only varies by 5%–7% given per °C change in temperature. Based on ERA5 reanalysis data, Kong and Huber (2024) demonstrated that the exceedance of T_{nw} over T_w at daytime is mostly within 3°C and centered around 1.2°C, and they are very close to each other at nighttime. Therefore, we expect similar changes in T_{nw} and T_w given that their differences in absolute values are not too large.

In summary, solar radiation drives T_g and T_{nw} away from T_a and T_w under the modulation of wind. The intricate parameterization of radiative and convective heat and/or mass transfer also complicates the intuitive understanding of T_g and T_{nw}, and therefore WBGT. However, in a simplified scenario with constant solar radiation and wind speed, changes in T_g and T_{nw}, and thus WBGT, can be interpreted similarly to changes in T_a and T_w.

2.4. Linearization

Equation 11 evaluates the response of WBGT to changes in common meteorological variables. However, as demonstrated in Section 2.2, all the sensitivity coefficients are state-dependent and vary across the multi-dimensional climatic phase space. The framework would be more practically useful if these coefficients can be approximated as constants without introducing considerable biases. This would allow WBGT to be expressed as a linear combination of T_a, q, P_s, and terms representing the effects of solar radiation and wind. This approach establishes a straightforward and mathematically tractable connection between WBGT and its meteorological inputs. Achieving this involves linearizing Equation 11 by computing the coefficients against a reference point, equivalent to retaining only the first-order term of a Taylor expansion of WBGT. Specifically, when solar radiation, wind speed and surface pressure remain unchanged, WBGT simplifies into a linear weighted sum of T_a and q, analogous to moist enthalpy.

The magnitude of biases introduced by linearization depends on two factors: the variability of the sensitivity coefficients across the multi-dimensional climatic phase space (i.e., the strength of nonlinearity), and the deviation from the reference point involved in the problem under consideration. To quantify these biases, we calculate pseudo values of T_g, T_{nw}, and WBGT (T_g^p, T_{nw}^p, WBGT^p) using the linearized framework.

$$T_g^p = \widehat{T_g} + \Delta T_a + \widehat{\alpha_{SR}} \Delta SR_g + \widehat{\alpha_h} \Delta h_{cg} \quad (20)$$

$$T_{nw}^p = \widehat{T_{nw}} + \widehat{\beta_{T_a}} \Delta T_a + \widehat{\beta_q} \Delta q + \widehat{\beta_{SR}} \Delta SR_w + \left(\widehat{\beta_\kappa} \Delta \kappa + \widehat{\beta_h} \Delta h_{cw}\right) + \widehat{\beta_{P_s}} \Delta P_s \quad (21)$$

$$WBGT^p = 0.7 T_{nw}^p + 0.2 P T_g^p + 0.1 T_a \quad (22)$$

which are then compared against the true values across a wide range of T_a, relative humidity (RH), wind speed, and solar radiation (Figures 2–4). In Equations 20 and 21, parameters with hats are calculated at a reference point with T_a of 32°C, RH of 50%, wind speed of 2 m/s, and solar radiation of $500 \mathrm{W/m^2}$ (marked as stars in Figures 2–4); the Δ terms represent deviations from this reference point. Surface pressure is consistently fixed at 1,000 hPa in Figures 2–4.

Biases for T_g are within ±1°C across the majority of the phase space (Figure 2). However, underestimations exceeding 4°C occur under substantially stronger solar radiation ($700 \mathrm{W/m^2}$) and weaker wind (0.5 m/s) than the reference values. In such scenarios, the magnitudes of both the sensitivity coefficients of solar radiation (α_{SR}) and wind (α_h) are considerably underestimated when calculated from the reference point. This underestimates the heating from a combination of increased solar radiation and less efficient convective heat loss as solar radiation gets stronger and wind gets weaker than the reference value, and therefore causes substantial negative biases.

Comparatively smaller negative biases also tend to occur as solar radiation gets weaker and wind gets stronger than the reference point. This is associated with the overestimation of α_{SR} and α_h by their reference values, and consequently an overestimation of the cooling from decreased solar radiation and more efficient convective heat loss. Positive biases under drier conditions are likely due to the more significant deviation of the air from the black body.

Biases for T_{nw} are primarily influenced by deviations in T_a and RH from the reference values, with less impact from wind and solar radiation anomalies (Figure 3). This indicates that linearizations to β_{T_a} and β_q are the primary sources of bias. As demonstrated in Section 2.2.2, both β_{T_a} and β_q are tight functions of T_{nw} (Figure 1d). Therefore, we expect smaller biases near the isoline of reference level T_{nw}, as confirmed in Figure 3. Note that T_{nw} and moist enthalpy share the same relative weights on T_a and q, and therefore have isolines of identical shape in a temperature-humidity phase space. As a result, we expect the linearized framework to perform well if the physical processes under consideration mainly involve relative partition between sensible and latent heat without changing moist enthalpy too much.

It is noteworthy that deviations in T_a and RH from their reference values while keeping solar radiation and wind speed at reference levels, consistently overestimates T_{nw} (Figure 3). This is linked to the simultaneous changes in T_{nw}, β_{T_a}, and β_q when moving across T_{nw} isolines. By approximating $\beta_q/\beta_{T_a} \approx L_v/C_p$, the components of T_a and q changes in Equation 21 can be rewritten as:

$$\widehat{\beta_{T_a}}\Delta T_a + \widehat{\beta_q}\Delta q = \widehat{\beta_{T_a}}/C_p(C_p\Delta T_a + L_v\Delta q)$$

For conditions with higher T_{nw} than the reference value, we have $C_p\Delta_{T_a} + L_v\Delta_q > 0$ and an overestimated β_{T_a} as calculated from the reference point. This exaggerates the resultant increases in T_{nw}. Conversely, β_{T_a} is underestimated when $C_p\Delta_{T_a} + L_v\Delta_q < 0$, which suppresses the T_{nw} decreases. In both cases, T_{nw} is overestimated.

We also explore the impact of surface pressure changes on T_{nw} biases. Since the effects of ΔP_s in the framework are conditioned on constant q (Equation 21), we reproduce Figure 3 within the T_a-q phase space for a surface pressure of 1000 hPa (Figure S1 in Supporting Information S1), 1050 hPa (Figure S2 in Supporting Information S1), 950 hPa (Figure S3 in Supporting Information S1) and 800 hPa (Figure S4 in Supporting Information S1). When fixing other parameters at the reference level, the effects of ΔP_s are approximately linear, and ±50hPa change from the reference value only vary T_{nw} by around ±0.6°C (Figure S5 in Supporting Information S1). Even a dramatic decrease in P_s from 1000 hPa to 800 hPa, typically corresponding to a strong increase in elevation, reduces T_{nw} by only 2.6°C (corresponding to a 1.8°C change in WBGT) (Figure S5 in Supporting Information S1). This is consistent with the small magnitudes and variability of β_{P_s} (Figure 1i). This small and linear impact of ΔP_s indicates small biases introduced from ΔP_s alone. Meanwhile, the sensitivity coefficients of other parameters (especially β_{T_a} and β_q) are also insensitive to ΔP_s because of its small influences on T_{nw} values. This suggests limited interaction between the effects of ΔP_s and changes in other parameters in causing T_{nw} biases. As a result, the T_{nw} bias structure and magnitudes remain almost identical across different levels of surface pressure (Figures S1–S4 in Supporting Information S1), indicating overall minimal biases introduced by changes in surface pressure.

WBGT biases follow the bias structure of T_{nw} with smaller magnitudes (Figure 4). In practice, the reference point can be selected properly in order to minimize biases. The evaluation conducted here provides confidence in the accuracy of the framework especially when temperature and humidity changes approximately follow moist enthalpy isolines. Building upon the idealized investigation above, we now demonstrate the strength of this framework when applied to more realistic and relevant settings.

3. Application of the Framework to Understanding the Driving Mechanism of WBGT Changes

According to the sensitivity framework, changes in WBGT consist of 30% of T_a changes and 70% of T_w changes (note that T_w is also affected by T_a as shown in Equation 18) when wind speed and solar radiation remain constant. The framework can be linearized transforming WBGT into a linear combination of its meteorological inputs. Therefore, if wind speed, solar radiation, and surface pressure are held constant, WBGT can be approximated as a linear weighted sum of T_a and q, analogous to moist enthalpy.

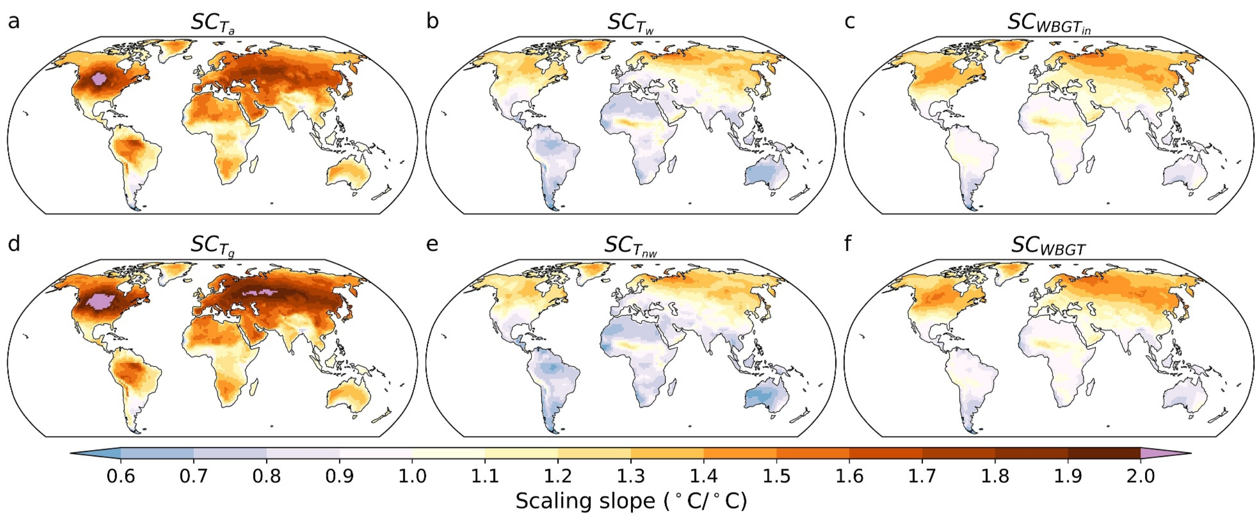

Figure 5. Scaling of summer average (a) T_a, (b) T_w, (c) $WBGT_{in}$, (d) T_g, (e) T_{nw} and (f) Wet-bulb globe temperature with global warming. $WBGT_{in}$ is calculated as $WBGT_{in} = 0.7T_w + 0.3T_a$.

The explicit connection between WBGT and T_a, q, T_w, moist enthalpy, etc., makes it possible to elucidate the atmospheric dynamic and thermodynamic control of WBGT variations by leveraging existing theories or methodologies developed for constraining these variables (Byrne & O'Gorman, 2013, 2016; Cronin, 2013; McColl & Tang, 2024). To illustrate this, here we apply the framework to two use scenarios: understanding the scaling of local summer average WBGT with global warming, and investigating the driving factors of extreme WBGT synoptic events. Although the ultimate goal is to understand the physical mechanisms, here we focus on the first step which is establishing an explicit connection between WBGT and these more intuitively understandable variables in the context of the problem. Further exploration into the physical mechanisms will be left for future studies.

3.1. Example Application I: Understanding the Scaling of WBGT With Global Warming

The scaling pattern of T_a, defined as the spatial distribution of how local T_a scales with global warming, has been well studied (Argüeso et al., 2016; Diffenbaugh & Giorgi, 2012; Orlowsky & Seneviratne, 2012; Suarez-Gutierrez et al., 2020). However, T_a—as a single dimension of the thermal environment—may not adequately capture the scaling pattern of heat stress, potentially misrepresenting hotspots most at risk of escalating heat stress (Matthews, 2018). Here, we briefly examine the scaling pattern of WBGT in comparison with that of T_a.

For demonstration purpose, WBGT is calculated at 3-hourly interval based on both historical and SSP585 simulations (covering the period 1960–2100) from a single CMIP6 climate model (ACCESS-CM2) (Dix et al., 2019). Scaling slopes are estimated by a linear regression of the summer (May–September for northern hemisphere and November–March for southern hemisphere) average of WBGT or other variables of interest to the annual global mean temperature over the 141-year period. We use SC_X to denote the scaling slope of any variable X.

We first confirm familiar features of the T_a scaling pattern as revealed in prior work (Brouillet & Joussaume, 2019; Diffenbaugh et al., 2007; Fan et al., 2020; Fischer & Schär, 2010; Meehl & Tebaldi, 2004; Vogel et al., 2020), such as regionally amplified warming over the eastern Europe and Central Asia, central North America, the Amazon, and the Sahara and Arabia deserts (Figure 5a). However, these hotspots no longer exist in case of $SC_{T_{nw}}$, and in fact become relative coldspots compared with other regions within similar latitudes (Figure 5e). The central Sahel emerges as a local hotspot of $SC_{T_{nw}}$. SC_{WBGT} shares similar spatial distribution and hotpots with $SC_{T_{nw}}$ (Figure 5f). The SC_{T_a} hotspots, although not becoming coldspots as in the case of $SC_{T_{nw}}$, do not manifest themselves in SC_{WBGT} as well.

To help understand factors controlling WBGT scaling, we decompose it into contributions from changes in T_a, q, wind, solar radiation and surface pressure using the linear sensitivity framework. Dividing the total differential of annual global mean temperature on both sides of Equation 11, we have

$$\begin{aligned} SC^{p}_{WBGT} = {} & \overbrace{\widehat{\gamma_{T_a}} SC_{T_a}}^{\text{Temperature}} + \overbrace{\widehat{\gamma_q} SC_q}^{\text{humidity}} + \overbrace{\left(0.7\widehat{\beta_{SR}} SC_{SR_v} + 0.2\widehat{\alpha_{SR}} SC_{SR_g}\right)}^{\text{Solar radiation}} \\ & + \underbrace{\left(0.7\widehat{\beta_\kappa} SC_\kappa + 0.7\widehat{\beta_h} SC_{h_{cw}} + 0.2\widehat{\alpha_h} SC_{h_{cg}}\right)}_{\text{Wind}} + \underbrace{0.7\widehat{\beta_{P_s}} SC_{P_s}}_{\text{Surface pressure}} \end{aligned} \tag{23}$$

where sensitivity coefficients with hats are calculated against historical (1960–2014) summer climatology within each grid cell. By doing so, the coefficients become constants locally but vary spatially depending on local climatology. This choice is motivated by the potentially large spatial variability in the coefficients due to substantial variations in climate from one location to another. Linearizing against a single location and applying it globally may introduce large biases. In comparison, temporal variations arising from climate change are expected to be a second order feature. Therefore, we opt to linearize against the historical climatology at each grid cell. Effectively, we account for the nonlinearity due to spatial variation while disregarding that from climate change. SC^{p}_{WBGT} indicates pseudo WBGT scaling generated from the framework.

We first validate the framework by showing that the scaling pattern of WBGT can be well captured by SC^{p}_{WBGT} (compare Figures 5f and 6a), despite minor overestimations that are smaller than 0.1°C over most of the world, and within 0.16°C globally (Figure 6b). These overestimations mainly arise from the linearization of β_{T_a} and β_q, since changes in other meteorological variables are of minor importance as demonstrated below. Both β_{T_a} and β_q decrease as T_{nw} rises (Figure 1d), and therefore are overestimated when calculated from historical climatology, whereas T_{nw} continues increasing into the future. This causes overestimations in $SC_{T_{nw}}$ and SC_{WBGT}, especially over the central Sahel and northern Siberia (Figure 6b), where T_{nw} is projected to rise faster (Figure 5e).

The biases could be reduced by calculating sensitivity coefficients against the climatology across the whole study period. However, this is not attempted here for several reasons. First, the biases are relatively small in most cases, and the scaling pattern is well captured. Namely, climate change has a relatively small influence on the values of sensitivity coefficients locally at each grid cell. Second, given $\beta_{T_a}/\beta_q \approx C_p/L_v$, any biases resulting from ignoring nonlinearities due to climate change will affect β_{T_a} and β_q in proportion, thus having a relatively small impact on the relative importance of changes in T_a and q. Finally, it is useful to demonstrate that it is sufficient to derive sensitivity coefficients from historical climatology, which is already available in observations or reanalysis data. This can be combined with model-projected changes in each meteorological input to provide a pseudo projection of WBGT changes with global warming.

As shown in Figure 6, SC_{WBGT} is predominantly influenced by T_a and q increases with minor contributions (within ±0.02°C/°C across the majority of the world) from changes in wind, solar radiation and surface pressure. T_a increases have a stronger effect than q enhancement in most regions except limited areas within the central Sahel, Indus Valley, Bangladesh, and equatorial east Africa (Figure 6e). However, changes in q may play an important role in shaping the spatial distribution of SC_{WBGT}, and causing the discrepancy between the scaling pattern of WBGT and T_a. For example, the regionally amplified warming over the aforementioned SC_{T_a} hotspots is counteracted by the regionally damped increases in q (Figure 6d). The Sahel hotspot in $SC_{T_{nw}}$ and SC_{WBGT} roots in a regionally amplified contributions from rising q (Figure 6d).

Understanding WBGT scaling is greatly simplified by the fact that changes in wind, solar radiation and surface pressure are of negligible importance (Figures 6f–6h) and therefore drop out of the problem. It suggests that scaling in T_g and T_{nw} should converge to scaling in T_a and T_w, as confirmed in Figure 5. Therefore, despite considerable discrepancies in absolute values, scaling in T_g and T_{nw} and the underlying physical control can be understood in the same way as for T_a and T_w. It also suggests that the frequently used indoor WBGT approximation—$WBGT_{in} = 0.7T_w + 0.3T_a$—although subject to considerable biases under

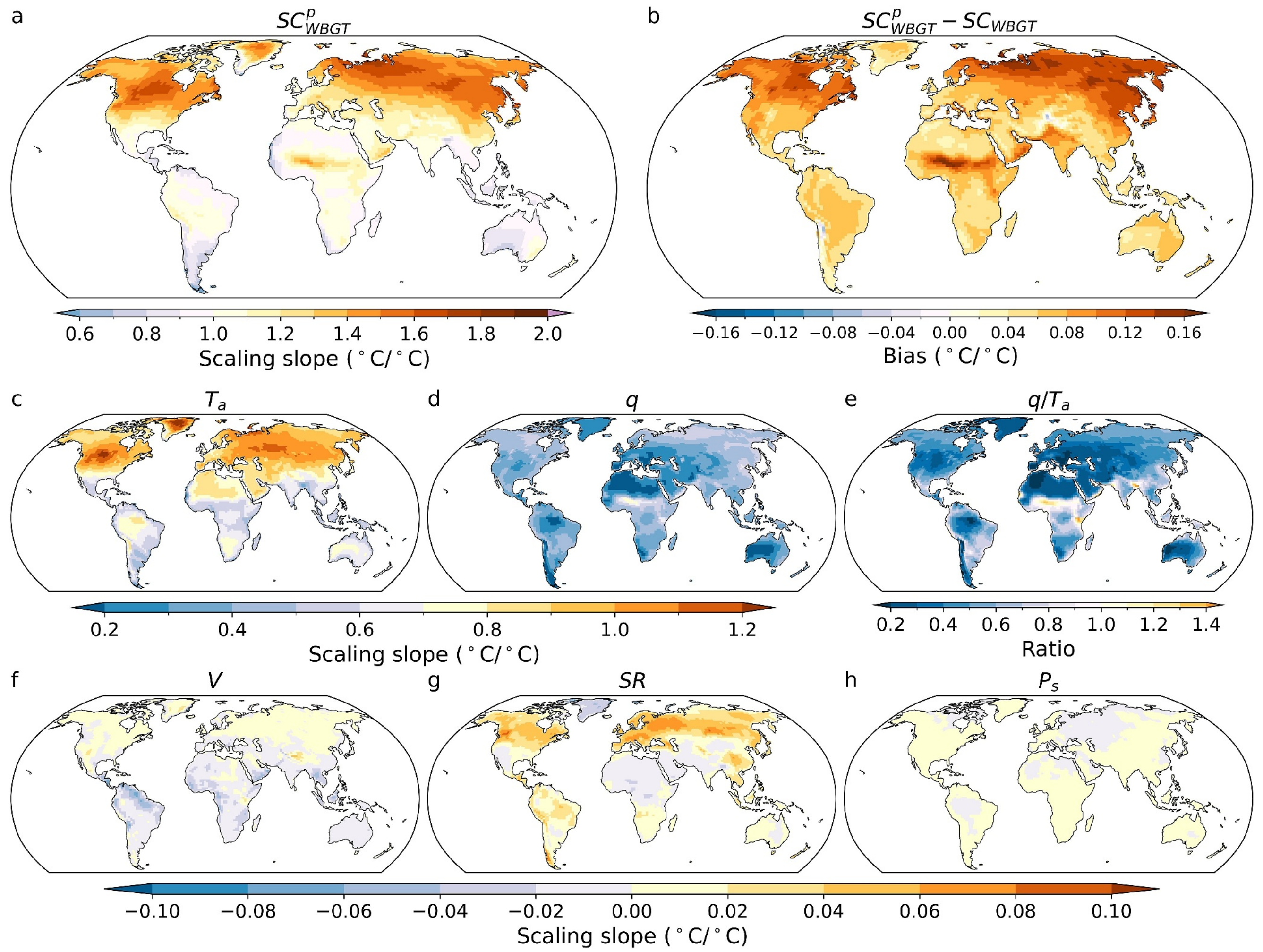

Figure 6. (a) *Pseudo*-scaling of summer average wet-bulb globe temperature (WBGT) derived from the framework, and (b) its biases in comparison with the true scaling. The relative contributions from changes in panels (c) T_a, (d) q, (f) wind speed, (g) solar radiation, and (h) surface pressure to WBGT scaling. Panel (e) shows the ratio between the effects of q and T_a changes.

outdoor conditions at daytime, may be sufficient for studying WBGT scaling with climate change (Figures 5c and 5f). In other words, *changes* to indoor WBGT are likely to be correct when applied to outdoor condition.

By comparing relative contributions from changes in different meteorological inputs, we can identify variables of particular importance in constraining heat stress projection, and get insights into the potential physical control of WBGT scaling across different regions. For example, the compensation between regionally amplified and damped increases in T_a and q over SC_{T_a} hotspots may come from a reallocation of surface energy from latent to sensible heat associated with the soil drying tendency over these regions (Qiao et al., 2023; Seneviratne et al., 2006; Vogel et al., 2017). In fact, damped q increases are able to outweigh amplified warming (Coffel et al., 2019), turning SC_{T_a} hotspots into $SC_{T_{nw}}$ coldspots. This is potentially owing to a deeper atmosphere boundary layer and stronger dry air entrainment from free troposphere under drier soil (Eltahir, 1998; Findell & Eltahir, 2003; Kong & Huber, 2023; Mishra et al., 2020; Pal & Eltahir, 2001). The Sahel hotspot of $SC_{T_{nw}}$ and SC_{WBGT}, dominated by a regionally amplified increases in q, is probably associated with changes in the West African Monsoon (Ramel et al., 2006; Roehrig et al., 2013). Given the predominant influences from T_a and q changes, the physical control of WBGT scaling can be quantitatively investigated by examining the scaling of physical quantities that constrain T_a and q variations, such as surface radiative and turbulent fluxes, temperature and moisture advection, soil moisture, precipitation and boundary layer depth.

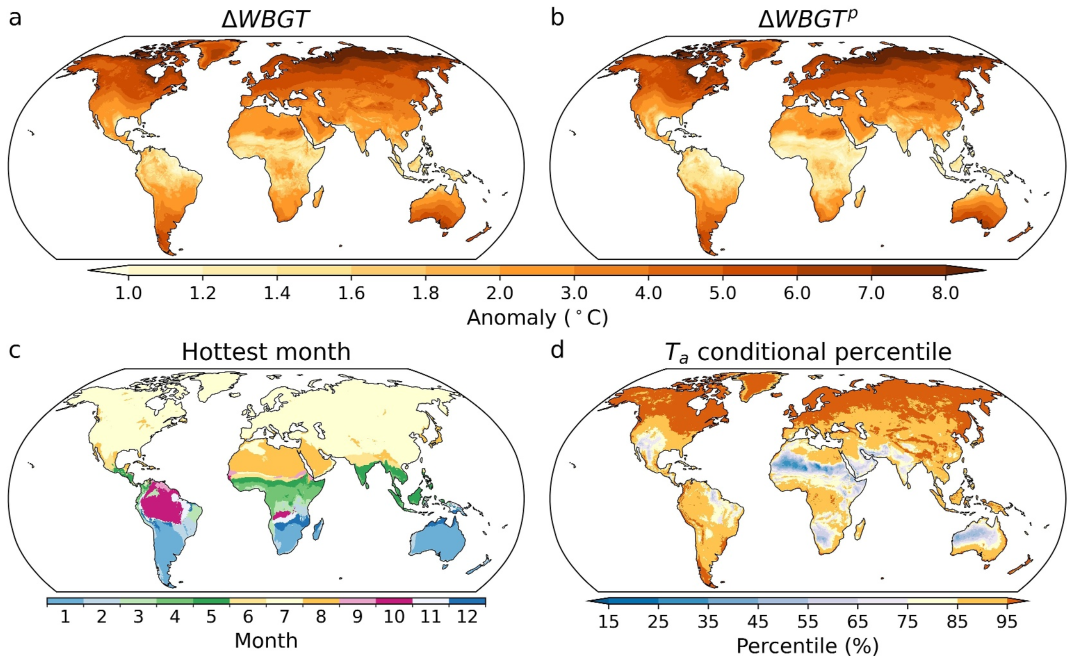

Figure 7. (a) Hot-day composite anomalies in wet-bulb globe temperature (WBGT) and (b) the pseudo anomalies generated from the framework. (c) The calendar month with hottest monthly average WBGT. (d) Average percentile of T_a conditioned on WBGT hot-days.

3.2. Example Application II: Exploring Drivers of Extreme WBGT Values

Understanding the physical drivers of extreme heat stress is important for prediction and mitigation (Domeisen et al., 2022; C. Ivanovich et al., 2022; Y. Zhang et al., 2024). WBGT is influenced by multiple environmental parameters, anomalies in any of which, or a combination thereof, may contribute to extreme WBGT values (Buzan & Huber, 2020). Depending on the relative roles of different parameters, extremely high WBGT values may manifest in diverse regimes, such as very hot but dry or moderately hot but humid, each controlled by potentially distinct mechanisms. This highlights the importance of identifying the prevailing regime of WBGT extremes over different regions in order to better understand the driving mechanism. This can be achieved by disentangling the relative roles of anomalies in different parameters in creating WBGT extremes using the linear sensitivity framework.

WBGT is calculated from ERA5 reanalysis data (Hersbach et al., 2018) at hourly interval and then aggregated to daily average scale during the period 1979–2022. Days with the daily average WBGT exceeding the local 95th percentile threshold are defined as WBGT hot-days (hereafter referred as hot-days for short). Both the threshold definition and hot-days identification are conditioned on the calendar month with highest monthly average WBGT across the study period (hereafter referred as hottest month) for each grid cell (Figure 7c). Hot-day composite anomalies in WBGT are calculated by comparing against the hottest month climatology, and can be decomposed into contributions from anomalies in different meteorological variables.

$$
\begin{aligned}
\Delta WBGT^p = {} & \overbrace{\widehat{\gamma_{T_a}}\Delta T}^{\text{Temperature}} + \overbrace{\widehat{\gamma_q}\Delta q}^{\text{humidity}} + \overbrace{\left(0.7\widehat{\beta_{SR}}\Delta SR_w + 0.2\widehat{\alpha_{SR}}\Delta SR_g\right)}^{\text{Solar radiation}} \\
& + \underbrace{\left(0.7\widehat{\beta_\kappa}\Delta\kappa + 0.7\widehat{\beta_h}\Delta h_{cw} + 0.2\widehat{\alpha_h}\Delta h_{cg}\right)}_{\text{Wind}} + \underbrace{0.7\widehat{\beta_{P_s}}\Delta P_s}_{\text{Surface pressure}}
\end{aligned}
\tag{24}
$$

where the sensitivity coefficients with hats are computed from the hottest month climatology within each grid cell; Δ terms represent hot-day composite anomalies, and $\Delta WBGT^p$ denote pseudo WBGT anomaly derived from the framework.

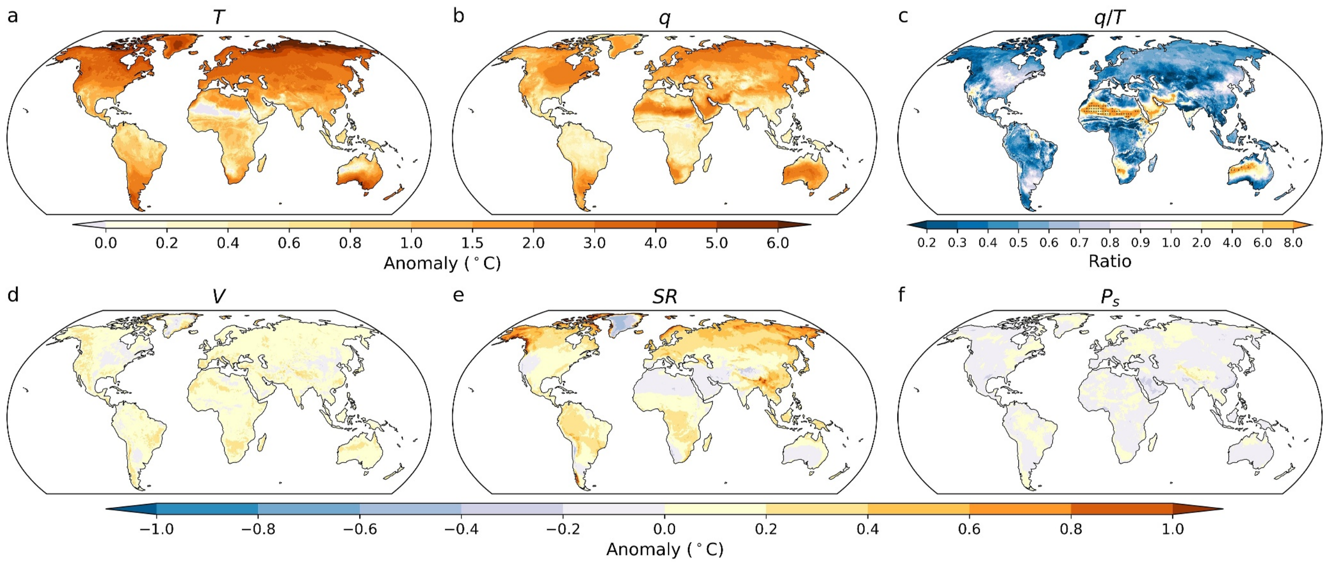

Figure 8. Relative contributions from anomalies in panels (a) T_a, (b) q, (d) wind speed, (e) solar radiation, and (f) surface pressure to hot-day composite anomalies in wet-bulb globe temperature. Panel c shows the ratio between the effects of q and T_a anomalies.

We first confirm that hot-day composite anomalies in WBGT can be well captured by $\Delta WBGT^p$ in terms of both magnitudes and spatial distribution (compare Figures 7a and 7b). Wind and surface pressure anomalies only have small effects mostly within ±0.2°C (Figures 8d and 8f). Solar radiation anomalies enhance $\Delta WBGT$ by less than 0.4°C in the tropics, and 0.2–0.8°C in the middle-high latitudes (Figure 8e). It suggests that WBGT hot-days in both regions are characterized by anomalous strong solar radiation corresponding to less cloud cover and potentially large-scale subsidence, as typically seen in high temperature days. This is consistent with the fact that evapotranspiration in both regions are "energy limited" climatologically (Seneviratne et al., 2010; Teuling et al., 2009), and solar radiation surplus enhances both surface latent and sensible heat flux, and therefore T_a and q.

Note that the increased evaporative demand from higher T_a could dry the soil and temporally shift these regions into the "moisture-limited" regime where soil moisture becomes the limiting factor for evapotranspiration (Seneviratne et al., 2006, 2010; Teuling et al., 2009). The drying soil reallocates radiative energy from latent to sensible heat flux and further raises T_a, constituting a positive feedback loop (Seneviratne et al., 2010). This may partially explain why Δq only contributes 20%–60% as ΔT_a in these regions (Figure 8c). In fact, the average percentiles of T_a conditional on WBGT hot-days exceed 85% over these regions, and even reach beyond 95% in the high-latitudes (Figure 7d). Namely, WBGT hot-days in the tropics and high latitudes largely overlap with high temperature days suggesting they are under the control of similar dynamic and thermodynamic processes. This is in contrast to the substantially lower overlapping ratio between T_w and T_a hot-days (<30% in the tropics and subtropics, and <70% in the high latitudes) (Kong & Huber, 2023).

In comparison, Δq outweighs the effects of ΔT_a over the Sahel, southeast Arabia Peninsular, the Indus Valley, southwest North America, south Africa, and central Australia (Figure 8c). Especially over the Sahel, $\Delta WBGT$ entirely comes from Δq with T_a cooler than average (Figures 8a and 8b). Therefore, we expect WBGT hot-days in these regions to be primarily controlled by processes that regulate humidity, such as sea breezes (Raymond et al., 2021, 2024), moisture convergence by the monsoon flow, and surface evapotranspiration (Monteiro & Caballero, 2019). These regions are characterized by "moisture limited" regime and have been identified as land-atmosphere coupling hotspots (Dirmeyer, 2011; Koster et al., 2004, 2006; Miralles et al., 2012; Schwingshackl et al., 2017). Wetter soil from precipitation surplus effectively reallocates surface radiative energy from sensible to latent heat flux. This suppresses boundary layer growth, traps surface enthalpy flux within a smaller volume, and reduces free-troposphere dry air entrainment into the boundary layer, which tends to amplify heat stress (Kong & Huber, 2023). These mechanisms have been suggested to be responsible for a positive coupling between soil moisture and T_w identified in these regions (Kong & Huber, 2023).

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Interestingly, Δq appears to be negative over parts of East Africa and central India (Figure 8b) where the hottest month is April or May and June (Figure 7c) corresponding to the climatologically wet local monsoon season. This suggests that WBGT hot-days over these regions during that time of the year are probably associated with monsoon breaks and dry spells with anomalously high temperature. Moving south, the negative Δq over central India suddenly becomes positive and makes larger contributions than ΔT_a (Figures 8b and 8c). This sharp transition aligns with a change of the hottest month from June to May (Figure 7c). It suggests that, during this climatologically drier pre-monsoon period, WBGT hot-days are probably associated with moisture advection from nearby ocean or earlier monsoon onset that brings anomalous high moisture. Similar dependence on the background wetness has also been found for the influence of precipitation on T_w over South Asia (C. C. Ivanovich et al., 2024).

As demonstrated above, the decomposition of WBGT anomalies allows us to identify key variables in creating WBGT extremes, leverage prior knowledge on the physical control of T_a and q variations, and formulate hypothesis on the driving mechanism. Since ΔT_a and Δq are of primary importance in most regions, one could investigate the governing processes of WBGT extremes through a formal budget analysis for T_a and q from either Lagrangian or Eulerian perspective (Li et al., 2015; Nieto et al., 2006; Röthlisberger & Papritz, 2023; Schumacher et al., 2019).

4. Comparison With the Simplified WBGT

As described in Kong and Huber (2022a), ad-hoc approximations of WBGT remain widely used (Buzan & Huber, 2020; de Lima et al., 2021; Kamal, Fahim, & Shahid, 2024; Kamal, Faruki Fahim, & Shahid, 2024; Y. Sun et al., 2024; Willett & Sherwood, 2010; Zhu et al., 2021). Some of these approximations were developed from empirical (linear or nonlinear) regressions of WBGT against multiple meteorological variables, based on a subset of samples across the entire multi-dimensional climatic phase space (ABM, 2010; Kamal, Faruki Fahim, & Shahid, 2024; Moran et al., 2001). These approximations share a similar formulation with the WBGT sensitivity framework presented here, except that the regression coefficients are empirically determined constants. It is useful to compare these ad-hoc approximations with our WBGT sensitivity framework to identify the similarities and discrepancies between both approaches and to highlight the advantages of our explicit and physics-based approach over empirical fitting across a limited sample.

Here we use the simplified WBGT ($sWBGT$) as an example due to its widespread use and linear formulation that is more analogous to our framework. $sWBGT$ was developed for preventing athlete heat injury (ACSM, 1984) and later adopted by the Australian Bureau of Meteorology (ABM, 2010). It only includes temperature and humidity while implicitly assuming moderately high radiation and light wind conditions.

$$sWBGT = 0.567(T_a - 273.15) + 0.00393e_a + 3.94 + 273.15$$

where both $sWBGT$ and T_a have units of K, and e_a has units of Pa. Employing $e_a \approx P_s q/\zeta$ and $\zeta = 0.622$, $sWBGT$ can be reformulated as

$$sWBGT = \eta_{T_a} T_a + \eta_q q + 122.2 \tag{25}$$

with $\eta_{T_a} = 0.567$ and $\eta_q = 0.00632P_s$.

Assuming constant surface pressure, $sWBGT$ becomes a linear weighted sum of T_a and q, similar to our linear sensitivity framework when solar radiation, wind, and surface pressure remain unchanged. Effectively, $sWBGT$ can be treated as a special case of the linear framework with sensitivity coefficients (empirically) determined under a specific set of reference condition (potentially a sunny and calm summer afternoon in the subtropics). When calculating $sWBGT$ globally, one applies the locally derived η_{T_a} and η_q to a wide range of conditions where the coefficients may strongly deviate from their reference values. This, in combination with the assumption of fixed solar radiation and wind speed, causes $sWBGT$ to have large systematic biases relative to WBGT (Budd, 2008; Grundstein & Cooper, 2018; Kong & Huber, 2022a; Qiu et al., 2024).

In comparison, the physics-based sensitivity framework in this study allows linearization against any reference point, and can be used to develop locally tuned linear approximations of WBGT suitable to any particular climate regime. The analytical form of sensitivity coefficients also aids in interpreting the sign and magnitude of biases when applying the linearized form beyond the reference condition, as demonstrated in example application I.

In fact, we can diagnose the source of biases in *sWBGT* using the framework. It is reasonable to assume that under the conditions where *sWBGT* was originally developed, this metric yields accurate estimates of WBGT, and the empirically determined η_{T_a} and η_q closely match theoretically derived values. We assume that these conditions correspond to a specific location in the world. Using this location as the reference point, *sWBGT* for any other place can be calculated by:

$$sWBGT^h = \widehat{WBGT} + \eta_{T_a}\Delta^s T + \eta_q \Delta^s q \tag{26}$$

where $\widehat{WBGT}$ denotes the reference level WBGT (equivalent to *sWBGT* in this case). The superscript h indicates that we are operating within a historical period, and Δ^s represents spatial changes within the historical period.

Equation 26 is expected to have substantial biases when applied to locations where T_{nw}—and consequently, the T_a and q sensitivity coefficients—deviate strongly from the reference values. Biases also arise from mismatches between actual solar radiation and wind speed and the fixed values implicitly assumed in *sWBGT*. An accurate estimate of WBGT requires correcting these biases

$$WBGT^h = \widehat{WBGT} + \left(\eta_{T_a} + \delta\eta_{T_a}\right)\Delta^s T + \left(\eta_q + \delta\eta_q\right)\Delta^s q + \Delta^s(S\&W) \tag{27}$$

where $\delta\eta_{T_a}$ and $\delta\eta_q$ are correction terms to both sensitivity coefficients; $\Delta^s(S\&W)$ represents the effects of changes in solar radiation and wind speed. Biases in *sWBGT* can be derived by subtracting Equation 27 from Equation 26

$$sWBGT^h - WBGT^h = -\left(\delta\eta_{T_a}\Delta^s T + \delta\eta_q \Delta^s q\right) - \Delta^s(S\&W) \tag{28}$$

Clearly, biases in *sWBGT* stem from two aspects: deviations of sensitivity coefficients from their reference values and the implicit assumption of fixed solar radiation and wind speed. As explained in Section 2.4, the first part typically leads to overestimation. Since *sWBGT* assumes moderately strong solar radiation and light wind, the second part is also more likely to cause overestimation, except under exceptionally strong solar radiation. Together, these factors explain the systematic overestimation by *sWBGT* (Figures 9a and 9b) (Grundstein & Cooper, 2018; Kong & Huber, 2022a; Qiu et al., 2024).

One may wonder whether the biases in *sWBGT* remain stationary with climate change, or in other words, whether *sWBGT* can reasonably capture changes in WBGT despite its tendency to overestimate absolute values. In Section 3.1, we demonstrate that it is sufficient to only account for changes in T_a and q to capture the scaling of summer average WBGT with global warming. Therefore, it is tempting to believe that *sWBGT* may be able to effectively capture WBGT scaling similar to the indoor WBGT approximation, since both approximations only consider temperature and humidity. To address this question, we modify Equations 26 and 27 to derive *sWBGT* and the accurate estimate of WBGT for a future warmer state.

$$sWBGT^f = \widehat{WBGT} + \eta_{T_a}\left(\Delta^s T + \Delta^t T\right) + \eta_q\left(\Delta^s q + \Delta^t q\right) \tag{29}$$

$$WBGT^f = \widehat{WBGT} + \left(\eta_{T_a} + \delta\eta_{T_a}\right)\left(\Delta^s T + \Delta^t T\right) + \left(\eta_q + \delta\eta_q\right)\left(\Delta^s q + \Delta^t q\right) + \Delta^s(S\&W) \tag{30}$$

where the superscript f signifies a future period, and Δ_t represents temporal changes from the historical to future period. In deriving Equation 30, we assume that the values of sensitivity coefficients and the effects of solar radiation and wind vary little with climate change. This is supported by the small biases introduced from calculating sensitivity coefficients against historical climatology and minor influences from changes in wind and solar radiation on WBGT scaling, as found in Section 3.1. Therefore, the sensitivity coefficients and the solar radiation and wind adjustment term remain identical between the historical and future periods (compare Equations 27 and 30).

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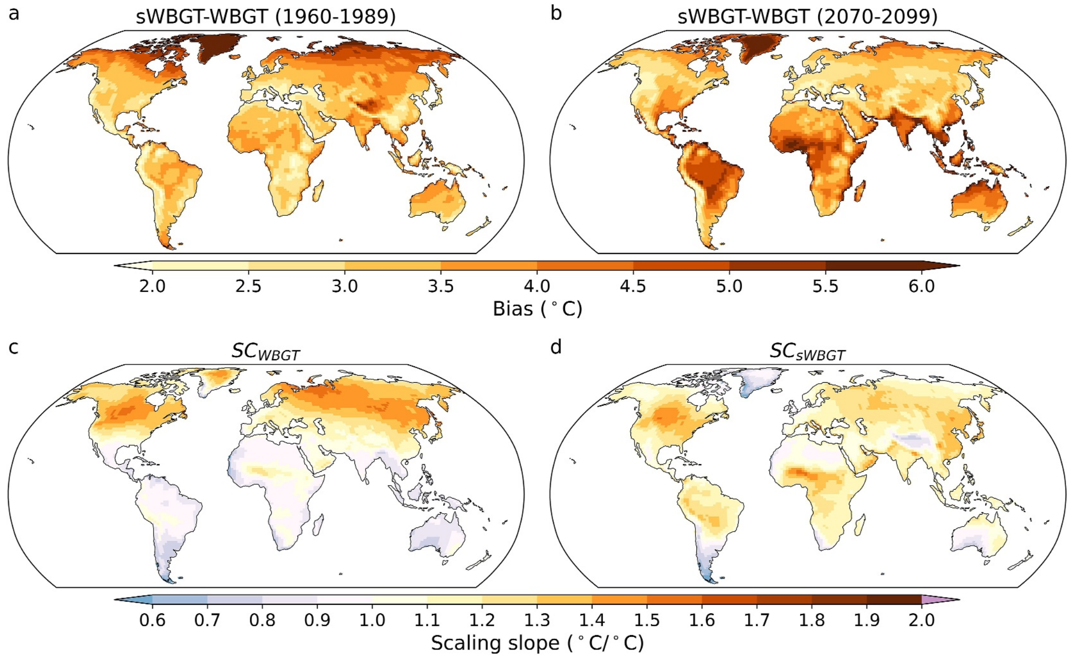

Figure 9. Biases in summer average wet-bulb globe temperature (WBGT) during (a) 1960–1989 and (b) 2070–2099. Scaling of summer average (c) WBGT and (d) *sWBGT* with global warming. The data set used for producing this figure comes from both historical and SSP585 simulations of ACCESS-CM2.

The biases in *sWBGT* for the future period are

$$sWBGT^f - WBGT^f = -\left[\delta\eta_{T_a}\left(\Delta^s T + \Delta^t T\right) - \delta\eta_q\left(\Delta^s q + \Delta^t q\right)\right] - \Delta^s(S\&W) \tag{31}$$

Comparing Equations 28 and 31, it is clear that, although the second part of the bias remains identical between historical and future periods, the first part is non-stationary. This will cause biases in the estimate of WBGT changes by *sWBGT*, which can be derived by subtracting Equation 28 from Equation 31.

$$\left(sWBGT^f - sWBGT^h\right) - \left(WBGT^f - WBGT^h\right) = -\delta\eta_{T_a}\Delta^t T - \delta\eta_q\Delta^t q \tag{32}$$

We can also anticipate the sign of the biases. For locations that are hotter and more humid, with T_{nw} higher than the reference level, both $\delta\eta_{T_a}$ and $\delta\eta_q$ are negative. Since $\Delta^t T$ and $\Delta^t q$ are typically positive, Equation 32 generates positive values, indicating amplified increases in WBGT. Conversely, WBGT increases tend to be underestimated in colder and drier regions, where T_{nw} is lower than the reference value.

To validate the arguments made above, we compare biases in the summer average *sWBGT* between a historical (1960–1989) and future period (2070–2099) (Figures 9a and 9b) using output from both historical and SSP585 simulations of ACCESS-CM2, and evaluate the scaling of the summer average *sWBGT* against the true WBGT scaling (Figures 9c and 9d) following the same approach as in example application I. We confirm that *sWBGT*'s overestimations are non-stationary, growing in hot-humid low latitudes and weakening in cold-dry high latitudes as climate warms (Figures 9a and 9b) (Qiu et al., 2024). This causes amplified and suppressed WBGT scaling in the respective regions, resulting in distinct scaling patterns between *sWBGT* and WBGT (Figures 9c and 9d).

The nonstationary biases from applying locally tuned sensitivity coefficients to distinctly different environmental conditions are expected to affect any linearly formulated heat stress metric. This is because these metrics approximate the nonlinear response of human heat strain to variations in different environmental parameters using locally calibrated linear relations.

5. Summary and Discussion

WBGT is the standard metric for workplace heat stress regulation and is widely used, although frequently in simplified forms, in climate change impact assessments (de Lima et al., 2021; Orlov et al., 2020; Saeed et al., 2022; Y. Sun et al., 2024). However, its application in understanding the atmospheric physical control of heat stress has been limited. This limitation stems from the complex parameterizations of radiative and convective heat and mass transfer embedded within the calculation of WBGT, which often makes this metric treated as a black box.

To address this challenge, we developed a sensitivity framework that analytically evaluates the response of WBGT to changes in its meteorological inputs. The sensitivity coefficients are state-dependent and vary across the multi-dimensional climatic phase space. A careful inspection of these coefficients reveals interactive effects between multiple meteorological variables. In a simplified scenario where wind speed and solar radiation remain constant, WBGT's two major components, T_{nw} and T_g, change at the same rate and direction as T_w and T_a, despite considerable differences in their absolute values.

Simpson et al. (2023) numerically derived the sensitivity of multiple heat stress metrics to temperature and humidity variations. Our framework instead analytically derive the sensitivity coefficients for WBGT. This helps establish an explicit connection between WBGT and common meteorological variables which might help remove the obscuring veil of WBGT's apparent complexity. In particular, the framework can be linearized by calculating sensitivity coefficients against a reference point. This transforms WBGT into a linear combination of T_a, q, surface pressure, and terms representing wind and solar radiation effects. If solar radiation, wind speed, and surface pressure remain unchanged, WBGT simplifies into a linear weighted sum of T_a and q, analogous to moist enthalpy.

These explicit and mathematically tractable relations between WBGT and more intuitively understandable variables enable us to understand the atmospheric dynamic and thermodynamic control of WBGT changes by leveraging existing theories and methodologies in understanding variations in these variables. To illustrate this, we apply the linear framework to disentangle the relative roles of T_a, q, and other meteorological variables in determining the scaling of WBGT locally with global warming and extreme WBGT events. This provides a solid foundation for understanding the underlying driving mechanisms.

WBGT is one of the most widely used heat stress metrics, with well-established safety thresholds to guide activity modifications in occupational (ACGIH, 2017; ISO, 2017; NIOSH, 2016), athletic (ACSM, 1984), and military settings (Army, 2003). It is also increasingly used for heat warning systems (Guo et al., 2024). However, WBGT is frequently applied in simplified forms, which can introduce substantial and less-transparent biases (Kong & Huber, 2022a). These biases may be large enough to shift values across severity categories, potentially altering critical decisions such as whether to issue heat warnings or not. Our linear framework offers a universal approach to developing locally tuned linear approximations of WBGT and diagnosing sources of biases in existing WBGT approximations. By enabling regionally tuned approximations and better awareness of these biases, the framework can enhance the reliability of WBGT-based safety regulations, improving their ability to protect public health.

There are caveats and potential limitations to the framework. The primary caveat is the potential biases introduced by linearization, when the actual values of sensitivity coefficients deviate from their reference values. The magnitudes of biases are expected to be small if the problem under consideration involves relatively small changes in moist enthalpy, and vice versa. Selecting an appropriate reference point is important to minimize these biases. The linear framework needs to be validated in practical applications due to these biases.

In addition, the sensitivity coefficients for WBGT concerning changes in wind speed and solar radiation are not directly assessed as sensitivities to per unit changes in these variables. Instead, they are formulated with respect to intermediate variables like mass/heat transfer efficiency and solar radiative heating, which themselves depend on wind speed and solar radiation. This approach simplifies the mathematical derivation of our framework. The direct sensitivity to wind speed and solar radiation changes, if needed, can be estimated by taking the derivatives of these intermediate variables with respect to wind speed and solar radiation.

Another caveat is the assumption of constant surface albedo when attributing changes in solar radiative heating (SR_g and SR_w) to changes in downward solar radiation. This assumption may not be valid in scenarios involving changes in land cover. In such cases, the effects of changes in albedo (a) and downward solar radiation (SR_{down}) can be further separated by expanding dSR_g and dSR_w with respect to both parameters (e.g., $dSR_g = (\partial SR_g/\partial a)\,da + (\partial SR_g/\partial SR_{down})\,dSR_{down}$).

Future work could explore various opportunities to integrate our framework with existing theories and methodologies to deepen the understanding of the physical control of WBGT changes. For example, combining our framework with the boundary layer climate sensitivity theory developed by Cronin (2013), one is able to analytically evaluate the response of WBGT to changes in land surface properties including canopy conductance, albedo, and aerodynamic roughness. The conceptual model of equal absolute changes in moist static energy and fractional changes in q between land and ocean can also be leveraged to predict WBGT changes over tropical land given the ocean warming (Byrne & O'Gorman, 2016; Y. Zhang & Fueglistaler, 2020; Y. Zhang et al., 2021). Our framework can also be combined with the two-resistance mechanism approach (Qin et al., 2023; Rigden & Li, 2017) or other similar methods (K. Zhang et al., 2023) to diagnose the physical control of urban-rural contrast in WBGT. Knowing the relative importance of T_a and q variations in determining WBGT changes, one could also apply temperature and moisture budget analysis to understanding the governing processes of WBGT changes.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

Hersbach et al. (2018) was downloaded from the Copernicus Climate Change Service (C3S) Climate Data Store (https://cds.climate.copernicus.eu/cdsapp#!/dataset/reanalysis-era5-single-levels?tab=form). The results contain modified Copernicus Climate Change Service information 2020. Neither the European Commission nor ECMWF is responsible for any use that may be made of the Copernicus information or data it contains. Dix et al. (2019) was downloaded from https://esgf-node.ipsl.upmc.fr/search/cmip6-ipsl/. The code for WBGT calculation is available at https://zenodo.org/record/5980536 (Kong & Huber, 2022b). The WBGT sensitivity framework is deposited at https://github.com/QINQINKONG/WBGT_sensitivity_framework.git (Kong & Huber, 2025).

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References

ABM. (2010). About the approximation to the WBGT used by the Bureau of Meteorology. Retrieved from http://www.bom.gov.au/info/thermal_stress/#approximation

ACGIH. (2017). *2017 tlvs and beis: Based on the documentation of the threshold limit values for chemical substances and physical agents and biological exposure indices* (Tech. Rep.). ACGIH.

ACSM. (1984). *Position stand on the prevention of thermal injuries during distance running* (Tech. Rep.). Medicine Sci. Sport. Exercise.

Argüeso, D., Di Luca, A., Perkins-Kirkpatrick, S. E., & Evans, J. P. (2016). Seasonal mean temperature changes control future heat waves. *Geophysical Research Letters*, *43*(14), 7653–7660. https://doi.org/10.1002/2016GL069408

Army, U. (2003). *Heat stress control and heat casualty management* (Tech. Rep.). Technical Bulletin Medical 507/Air Force Pamphlet.

Bröde, P., Fiala, D., Błażejczyk, K., Holmér, I., Jendritzky, G., Kampmann, B., et al. (2012). Deriving the operational procedure for the Universal Thermal Climate Index (UTCI). *International Journal of Biometeorology*, *56*(3), 481–494. https://doi.org/10.1007/s00484-011-0454-1

Brouillet, A., & Joussaume, S. (2019). Investigating the role of the relative humidity in the co-occurrence of temperature and heat stress extremes in CMIP5 projections. *Geophysical Research Letters*, *46*(20), 11435–11443. https://doi.org/10.1029/2019GL084156

Budd, G. M. (2008). Wet-bulb globe temperature (WBGT)—Its history and its limitations. *Journal of Science and Medicine in Sport*, *11*(1), 20–32. https://doi.org/10.1016/j.jsams.2007.07.003

Burke, M., González, F., Baylis, P., Heft-Neal, S., Baysan, C., Basu, S., & Hsiang, S. (2018). Higher temperatures increase suicide rates in the United States and Mexico. *Nature Climate Change*, *8*(8), 723–729. https://doi.org/10.1038/s41558-018-0222-x

Burke, M., Hsiang, S. M., & Miguel, E. (2015). Global non-linear effect of temperature on economic production. *Nature*, *527*(7577), 235–239. https://doi.org/10.1038/nature15725

Buzan, J. R., & Huber, M. (2020). Moist heat stress on a hotter Earth. *Annual Review of Earth and Planetary Sciences*, *48*(1), 623–655. https://doi.org/10.1146/annurev-earth-053018-060100

Byrne, M. P., & O'Gorman, P. A. (2013). Land–Ocean warming contrast over a wide range of climates: Convective quasi-equilibrium theory and idealized simulations. *Journal of Climate*, *26*(12), 4000–4016. https://doi.org/10.1175/JCLI-D-12-00262.1

Byrne, M. P., & O'Gorman, P. A. (2016). Understanding decreases in land relative humidity with global warming: Conceptual model and GCM simulations. *Journal of Climate*, *29*(24), 9045–9061. https://doi.org/10.1175/JCLI-D-16-0351.1

Cane, M. A., Miguel, E., Burke, M., Hsiang, S. M., Lobell, D. B., Meng, K. C., & Satyanath, S. (2014). Temperature and violence. *Nature Climate Change*, *4*(4), 234–235. https://doi.org/10.1038/nclimate2171

Chakraborty, T., Qian, Y., Li, J., Leung, L. R., & Sarangi, C. (2025). Daytime urban heat stress in North America reduced by irrigation. *Nature Geoscience*, *18*(1), 57–64. https://doi.org/10.1038/s41561-024-01613-z

Chilton, T. H., & Colburn, A. P. (1934). Mass transfer (absorption) coefficients prediction from data on heat transfer and fluid friction. *Industrial & Engineering Chemistry*, *26*(11), 1183–1187. https://doi.org/10.1021/ie50299a012

Coffel, E. D., Horton, R. M., Winter, J. M., & Mankin, J. S. (2019). Nonlinear increases in extreme temperatures paradoxically dampen increases in extreme humid-heat. *Environmental Research Letters*, *14*(8), 084003. https://doi.org/10.1088/1748-9326/ab28b7
Cronin, T. W. (2013). A sensitivity theory for the equilibrium boundary layer over land. *Journal of Advances in Modeling Earth Systems*, *5*(4), 764–784. https://doi.org/10.1002/jame.20048
d'Ambrosio Alfano, F. R., Palella, B. I., & Riccio, G. (2012). On the problems related to natural wet bulb temperature indirect evaluation for the assessment of hot thermal environments by means of WBGT. *Annals of Occupational Hygiene*, *56*(9), 1063–1079. https://doi.org/10.1093/annhyg/mes036
de Lima, C. Z., Buzan, J. R., Moore, F. C., Baldos, U. L. C., Huber, M., & Hertel, T. W. (2021). Heat stress on agricultural workers exacerbates crop impacts of climate change. *Environmental Research Letters*, *16*(4), 044020. https://doi.org/10.1088/1748-9326/abeb9f
Diffenbaugh, N. S., & Giorgi, F. (2012). Climate change hotspots in the CMIP5 global climate model ensemble. *Climatic Change*, *114*(3–4), 813–822. https://doi.org/10.1007/s10584-012-0570-x
Diffenbaugh, N. S., Pal, J. S., Giorgi, F., & Gao, X. (2007). Heat stress intensification in the Mediterranean climate change hotspot. *Geophysical Research Letters*, *34*(11), 2007GL030000. https://doi.org/10.1029/2007GL030000
Dirmeyer, P. A. (2011). The terrestrial segment of soil moisture-climate coupling. *Geophysical Research Letters*, *38*(16), L16702. https://doi.org/10.1029/2011GL048268
Dix, M., Bi, D., Dobrohotoff, P., Fiedler, R., Harman, I., Law, R., et al. (2019). CSIRO-ARCCSS ACCESS-CM2 model output prepared for CMIP6 ScenarioMIP ssp585 [Dataset]. *Earth System Grid Federation*. https://doi.org/10.22033/ESGF/CMIP6.4332
Domeisen, D. I. V., Eltahir, E. A. B., Fischer, E. M., Knutti, R., Perkins-Kirkpatrick, S. E., Schär, C., et al. (2022). Prediction and projection of heatwaves. *Nature Reviews Earth & Environment*, *4*(1), 36–50. https://doi.org/10.1038/s43017-022-00371-z
Ebi, K. L., Capon, A., Berry, P., Broderick, C., De Dear, R., Havenith, G., et al. (2021). Hot weather and heat extremes: Health risks. *The Lancet*, *398*(10301), 698–708. https://doi.org/10.1016/S0140-6736(21)01208-3
Eltahir, E. A. B. (1998). A soil moisture-rainfall feedback mechanism: 1. Theory and observations. *Water Resources Research*, *34*(4), 765–776. https://doi.org/10.1029/97WR03499
Fan, X., Duan, Q., Shen, C., Wu, Y., & Xing, C. (2020). Global surface air temperatures in CMIP6: Historical performance and future changes. *Environmental Research Letters*, *15*(10), 104056. https://doi.org/10.1088/1748-9326/abb051
Findell, K. L., & Eltahir, E. A. B. (2003). Atmospheric controls on soil moisture–boundary layer interactions. Part I: Framework development. *Journal of Hydrometeorology*, *4*(3), 552–569. https://doi.org/10.1175/1525-7541(2003)004⟨0552:ACOSML⟩2.0.CO;2
Fischer, E. M., & Schär, C. (2010). Consistent geographical patterns of changes in high-impact European heatwaves. *Nature Geoscience*, *3*(6), 398–403. https://doi.org/10.1038/ngeo866
Foster, J., Smallcombe, J. W., Hodder, S., Jay, O., Flouris, A. D., & Havenith, G. (2022). Quantifying the impact of heat on human physical work capacity; part II: The observed interaction of air velocity with temperature, humidity, sweat rate, and clothing is not captured by most heat stress indices. *International Journal of Biometeorology*, *66*(3), 507–520. https://doi.org/10.1007/s00484-021-02212-y
Foster, J., Smallcombe, J. W., Hodder, S., Jay, O., Flouris, A. D., Nybo, L., & Havenith, G. (2021). An advanced empirical model for quantifying the impact of heat and climate change on human physical work capacity. *International Journal of Biometeorology*, *65*(7), 1215–1229. https://doi.org/10.1007/s00484-021-02105-0
Foster, J., Smallcombe, J. W., Hodder, S., Jay, O., Flouris, A. D., Nybo, L., & Havenith, G. (2022). Quantifying the impact of heat on human physical work capacity; part III: The impact of solar radiation varies with air temperature, humidity, and clothing coverage. *International Journal of Biometeorology*, *66*(1), 175–188. https://doi.org/10.1007/s00484-021-02205-x
Gagge, A. P., & Gonzalez, R. R. (1996). Mechanisms of heat exchange: Biophysics and physiology. In R. Terjung (Ed.), *Comprehensive physiology* (1st ed., pp. 45–84). Wiley. https://doi.org/10.1002/cphy.cp040104
Grundstein, A., & Cooper, E. (2018). Assessment of the Australian Bureau of Meteorology wet bulb globe temperature model using weather station data. *International Journal of Biometeorology*, *62*(12), 2205–2213. https://doi.org/10.1007/s00484-018-1624-1
Guo, Q., Yuan, L., Ng, C. F. S., & Hashizume, M. (2024). Evaluating Japan's revised heat-health warning system in the face of recent escalating heat stress. *Environmental Research Letters*, *19*(5), 054002. https://doi.org/10.1088/1748-9326/ad3a81
Guo, Q., Zhou, X., Satoh, Y., & Oki, T. (2022). Irrigated cropland expansion exacerbates the urban moist heat stress in northern India. *Environmental Research Letters*, *17*(5), 054013. https://doi.org/10.1088/1748-9326/ac64b6
Havenith, G., Smallcombe, J. W., Hodder, S., Jay, O., & Foster, J. (2024). Comparing efficacy of different climate indices for predicting labor loss, body temperature, and thermal perception in a wide variety of warm and hot climates. *Journal of Applied Physiology*, *137*(2), 312–328. https://doi.org/10.1152/japplphysiol.00613.2023
Hersbach, H., Bell, B., Berrisford, P., Biavati, G., Horányi, A., Muñoz Sabater, J., et al. (2018). ERA5 hourly data on single levels from 1979 to present [dataset]. *Copernicus Climate Change Service (C3S) Climate Data Store (CDS)*. https://doi.org/10.24381/cds.adbb2d47
Hsiang, S. M., Burke, M., & Miguel, E. (2013). Quantifying the influence of climate on human conflict. *Science*, *341*(6151), 1235367. https://doi.org/10.1126/science.1235367
Ioannou, L. G., Tsoutsoubi, L., Mantzios, K., Vliora, M., Nintou, E., Piil, J. F., et al. (2022). Indicators to assess physiological heat strain – Part 3: Multi-country field evaluation and consensus recommendations. *Temperature*, *9*(3), 274–291. https://doi.org/10.1080/23328940.2022.2044739
ISO. (2017). *Ergonomics of the thermal environment — Assessment of heat stress using the WBGT (wet bulb globe temperature) index (International Standard)*. International Organization for Standardization (ISO).
Ivanovich, C., Anderson, W., Horton, R., Raymond, C., & Sobel, A. (2022). The influence of intraseasonal oscillations on humid heat in the Persian Gulf and South Asia. *Journal of Climate*, *35*(13), 1–48. https://doi.org/10.1175/JCLI-D-21-0488.1
Ivanovich, C. C., Horton, R. M., Sobel, A. H., & Singh, D. (2024). Subseasonal variability of humid heat during the south Asian summer monsoon. *Geophysical Research Letters*, *51*(6), e2023GL107382. https://doi.org/10.1029/2023GL107382
Kamal, A. S. M. M., Fahim, A. K. F., & Shahid, S. (2024). Changes in wet bulb globe temperature and risk to heat-related hazards in Bangladesh. *Scientific Reports*, *14*(1), 10417. https://doi.org/10.1038/s41598-024-61138-8
Kamal, A. S. M. M., Faruki Fahim, A. K., & Shahid, S. (2024). Simplified equations for wet bulb globe temperature estimation in Bangladesh. *International Journal of Climatology*, *44*(5), 1636–1653. https://doi.org/10.1002/joc.8402
Kjellstrom, T., Briggs, D., Freyberg, C., Lemke, B., Otto, M., & Hyatt, O. (2016). Heat, human performance, and occupational health: A key issue for the assessment of global climate change impacts. *Annual Review of Public Health*, *37*(1), 97–112. https://doi.org/10.1146/annurev-publhealth-032315-021740
Kong, Q., & Huber, M. (2022a). Explicit calculations of Wet Bulb Globe Temperature compared with approximations and why it matters for labor productivity. *Earth's Future*, *10*(3). https://doi.org/10.1029/2021EF002334

Kong, Q., & Huber, M. (2022b). Explicit calculations of wet bulb globe temperature compared with approximations and why it matters for labor productivity [software]. *Zenodo*, *10*(3). https://doi.org/10.5281/zenodo.5980536

Kong, Q., & Huber, M. (2023). Regimes of soil moisture-wet bulb temperature coupling with relevance to moist heat stress. *Journal of Climate*, *36*(22), 1–45. https://doi.org/10.1175/JCLI-D-23-0132.1

Kong, Q., & Huber, M. (2024). A new, zero-iteration analytic implementation of wet-bulb globe temperature: Development, validation, and comparison with other methods. *GeoHealth*, *8*(10), e2024GH001068. https://doi.org/10.1029/2024GH001068

Kong, Q., & Huber, M. (2025). A linear sensitivity framework for wet-bulb globe temperature [software]. *Zenodo*. https://doi.org/10.5281/zenodo.14728661

Koster, R. D., Dirmeyer, P. A., Guo, Z., Bonan, G., Chan, E., Cox, P., et al. (2004). Regions of strong coupling between soil moisture and precipitation. *Science*, *305*(5687), 1138–1140. https://doi.org/10.1126/science.1100217

Koster, R. D., Sud, Y. C., Guo, Z., Dirmeyer, P. A., Bonan, G., Oleson, K. W., et al. (2006). Glace: The global land–atmosphere coupling experiment. Part I: Overview. *Journal of Hydrometeorology*, *7*(4), 590–610. https://doi.org/10.1175/JHM510.1

Krakauer, N. Y., Cook, B. I., & Puma, M. J. (2020). Effect of irrigation on humid heat extremes. *Environmental Research Letters*, *15*(9), 094010. https://doi.org/10.1088/1748-9326/ab9ecf

Lemke, B., & Kjellstrom, T. (2012). Calculating workplace WBGT from meteorological data: A tool for climate change assessment. *Industrial Health*, *50*(4), 267–278. https://doi.org/10.2486/indhealth.MS1352

Lewis, W. K. (1922). The evaporation of a liquid into a gas. *Transactions of the American Society of Mechanical Engineers*, *44*, 325–332. https://doi.org/10.1115/1.4058175

Li, D., Sun, T., Liu, M., Yang, L., Wang, L., & Gao, Z. (2015). Contrasting responses of urban and rural surface energy budgets to heat waves explain synergies between urban heat islands and heat waves. *Environmental Research Letters*, *10*(5), 054009. https://doi.org/10.1088/1748-9326/10/5/054009

Liljegren, J. C., Carhart, R. A., Lawday, P., Tschopp, S., & Sharp, R. (2008). Modeling the wet bulb globe temperature using standard meteorological measurements. *Journal of Occupational and Environmental Hygiene*, *5*(10), 645–655. https://doi.org/10.1080/15459620802310770

Lu, Y.-C., & Romps, D. (2023a). Wet-bulb temperature or heat index: Which better predicts fatal heat in a warming climate? *Physiology*, *38*(S1), 5734524. https://doi.org/10.1152/physiol.2023.38.S1.5734524

Lu, Y.-C., & Romps, D. M. (2022). Extending the heat index. *Journal of Applied Meteorology and Climatology*, *61*(10), 1367–1383. https://doi.org/10.1175/JAMC-D-22-0021.1

Lu, Y.-C., & Romps, D. M. (2023b). Is a wet-bulb temperature of 35°C the correct threshold for human survivability? *Environmental Research Letters*, *18*(9), 094021. https://doi.org/10.1088/1748-9326/ace83c

Matthews, T. (2018). Humid heat and climate change. *Progress in Physical Geography: Earth and Environment*, *42*(3), 391–405. https://doi.org/10.1177/0309133318776490

McColl, K. A., & Tang, L. I. (2024). An analytic theory of near-surface relative humidity over land. *Journal of Climate*, *37*(4), 1213–1230. https://doi.org/10.1175/JCLI-D-23-0342.1

Meehl, G. A., & Tebaldi, C. (2004). More intense, more frequent, and longer lasting heat waves in the 21st century. *Science*, *305*(5686), 994–997. https://doi.org/10.1126/science.1098704

Miralles, D. G., van den Berg, M. J., Teuling, A. J., & de Jeu, R. A. M. (2012). Soil moisture-temperature coupling: A multiscale observational analysis: Soil moisture-temperature coupling. *Geophysical Research Letters*, *39*(21). https://doi.org/10.1029/2012GL053703

Mishra, V., Ambika, A. K., Asoka, A., Aadhar, S., Buzan, J., Kumar, R., & Huber, M. (2020). Moist heat stress extremes in India enhanced by irrigation. *Nature Geoscience*, *13*(11), 722–728. https://doi.org/10.1038/s41561-020-00650-8

Monteiro, J. M., & Caballero, R. (2019). Characterization of extreme wet-bulb temperature events in southern Pakistan. *Geophysical Research Letters*, *46*(17–18), 10659–10668. https://doi.org/10.1029/2019GL084711

Moran, D., Pandolf, K., Shapiro, Y., Heled, Y., Shani, Y., Mathew, W., & Gonzalez, R. (2001). An environmental stress index (ESI) as a substitute for the wet bulb globe temperature (WBGT). *Journal of Thermal Biology*, *26*(4–5), 427–431. https://doi.org/10.1016/S0306-4565(01)00055-9

Nieto, R., Gimeno, L., & Trigo, R. M. (2006). A Lagrangian identification of major sources of Sahel moisture. *Geophysical Research Letters*, *33*(18), 2006GL027232. https://doi.org/10.1029/2006GL027232

NIOSH. (2016). *Criteria for a recommended standard: Occupational exposure to heat and hot environments* (Tech. Rep. No. DHHS (NIOSH) Publication No. 2016-106). DHHS, NIOSH.

Orlov, A., Sillmann, J., Aunan, K., Kjellstrom, T., & Aaheim, A. (2020). Economic costs of heat-induced reductions in worker productivity due to global warming. *Global Environmental Change*, *63*, 102087. https://doi.org/10.1016/j.gloenvcha.2020.102087

Orlowsky, B., & Seneviratne, S. I. (2012). Global changes in extreme events: Regional and seasonal dimension. *Climatic Change*, *110*(3–4), 669–696. https://doi.org/10.1007/s10584-011-0122-9

Pal, J. S., & Eltahir, E. A. B. (2001). Pathways relating soil moisture conditions to future summer rainfall within a model of the land–atmosphere system. *Journal of Climate*, *14*(6), 1227–1242. https://doi.org/10.1175/1520-0442(2001)014⟨1227:PRSMCT⟩2.0.CO;2

Parajuli, S. P., Biggs, T., De Sales, F., Zavala Perez, M. A., He, C., Jones, C., et al. (2024). Impact of irrigation on farmworker's heat stress in California differs by season and during the day and night. *Communications Earth & Environment*, *5*(1), 787. https://doi.org/10.1038/s43247-024-01959-7

Parsons, K. C. (2014). *Human thermal environments: The effects of hot, moderate, and cold environments on human health, comfort, and performance* (3rd ed.). CRC Press, Taylor & Francis Group.

Perkins, S. E. (2015). A review on the scientific understanding of heatwaves—Their measurement, driving mechanisms, and changes at the global scale. *Atmospheric Research*, *164–165*, 242–267. https://doi.org/10.1016/j.atmosres.2015.05.014

Qiao, L., Zuo, Z., Zhang, R., Piao, S., Xiao, D., & Zhang, K. (2023). Soil moisture–atmosphere coupling accelerates global warming. *Nature Communications*, *14*(1), 4908. https://doi.org/10.1038/s41467-023-40641-y

Qin, Y., Liao, W., & Li, D. (2023). Attributing the urban–rural contrast of heat stress simulated by a global model. *Journal of Climate*, *36*(6), 1805–1822. https://doi.org/10.1175/JCLI-D-22-0436.1

Qiu, L., Zhu, Z., Zhou, Z., Im, E.-S., Min, S.-K., Kim, Y.-H., et al. (2024). Amplification of the discrepancy between simplified and physics-based wet-bulb globe temperatures in a warmer climate. *Weather and Climate Extremes*, *44*, 100677. https://doi.org/10.1016/j.wace.2024.100677

Ramel, R., Gallée, H., & Messager, C. (2006). On the northward shift of the West African monsoon. *Climate Dynamics*, *26*(4), 429–440. https://doi.org/10.1007/s00382-005-0093-5

Raymond, C., Matthews, T., Horton, R. M., Fischer, E. M., Fueglistaler, S., Ivanovich, C., et al. (2021). On the controlling factors for globally extreme humid heat. *Geophysical Research Letters*, *48*(23). https://doi.org/10.1029/2021GL096082

Raymond, C., Matthews, T., & Tuholske, C. (2024). Post-sunset humid-heat maxima near the southern Persian/Arabian Gulf. https://doi.org/10.21203/rs.3.rs-3919308/v1

Rigden, A. J., & Li, D. (2017). Attribution of surface temperature anomalies induced by land use and land cover changes. *Geophysical Research Letters*, *44*(13), 6814–6822. https://doi.org/10.1002/2017GL073811
Roehrig, R., Bouniol, D., Guichard, F., Hourdin, F., & Redelsperger, J.-L. (2013). The present and future of the West African monsoon: A process-oriented assessment of CMIP5 simulations along the amma transect. *Journal of Climate*, *26*(17), 6471–6505. https://doi.org/10.1175/JCLI-D-12-00505.1
Rothfusz, L. P. (1990). *The heat index equation (or, more than you ever wanted to know about heat index)* (Tech. Rep. No. SR 90-23). National Oceanic and Atmospheric Administration, National Weather Service, Office of Meteorology.
Röthlisberger, M., & Papritz, L. (2023). Quantifying the physical processes leading to atmospheric hot extremes at a global scale. *Nature Geoscience*, *16*(3), 210–216. https://doi.org/10.1038/s41561-023-01126-1
Saeed, W., Haqiqi, I., Kong, Q., Huber, M., Buzan, J. R., Chonabayashi, S., et al. (2022). The poverty impacts of labor heat stress in west Africa under a warming climate. *Earth's Future*, *10*(11). https://doi.org/10.1029/2022EF002777
Schumacher, D. L., Keune, J., van Heerwaarden, C. C., Vilà-Guerau de Arellano, J., Teuling, A. J., & Miralles, D. G. (2019). Amplification of mega-heatwaves through heat torrents fuelled by upwind drought. *Nature Geoscience*, *12*(9), 712–717. https://doi.org/10.1038/s41561-019-0431-6
Schwingshackl, C., Hirschi, M., & Seneviratne, S. I. (2017). Quantifying spatiotemporal variations of soil moisture control on surface energy balance and near-surface air temperature. *Journal of Climate*, *30*(18), 7105–7124. https://doi.org/10.1175/JCLI-D-16-0727.1
Seneviratne, S. I., Corti, T., Davin, E. L., Hirschi, M., Jaeger, E. B., Lehner, I., et al. (2010). Investigating soil moisture–climate interactions in a changing climate: A review. *Earth-Science Reviews*, *99*(3–4), 125–161. https://doi.org/10.1016/j.earscirev.2010.02.004
Seneviratne, S. I., Lüthi, D., Litschi, M., & Schär, C. (2006). Land–atmosphere coupling and climate change in Europe. *Nature*, *443*(7108), 205–209. https://doi.org/10.1038/nature05095
Simpson, C. H., Brousse, O., Ebi, K. L., & Heaviside, C. (2023). Commonly used indices disagree about the effect of moisture on heat stress. *npj Climate and Atmospheric Science*, *6*(1), 78. https://doi.org/10.1038/s41612-023-00408-0
Souri, A. H., Wang, H., González Abad, G., Liu, X., & Chance, K. (2020). Quantifying the impact of excess moisture from transpiration from crops on an extreme heat wave event in the midwestern U.S.: A top-down constraint from moderate resolution imaging spectroradiometer water vapor retrieval. *Journal of Geophysical Research: Atmospheres*, *125*(7), e2019JD031941. https://doi.org/10.1029/2019JD031941
Steadman, R. G. (1979). The assessment of sultriness. Part I: A temperature-humidity index based on human physiology and clothing science. *Journal of Applied Meteorology*, *18*(7), 861–873. https://doi.org/10.1175/1520-0450(1979)018⟨0861:TAOSPI⟩2.0.CO;2
Suarez-Gutierrez, L., Müller, W. A., Li, C., & Marotzke, J. (2020). Hotspots of extreme heat under global warming. *Climate Dynamics*, *55*(3–4), 429–447. https://doi.org/10.1007/s00382-020-05263-w
Sun, S., Zhang, Q., Shi, C., Singh, V. P., Zhang, T., Gu, J., et al. (2024). Urban irrigation reduces moist heat stress in Beijing, China. *npj Climate and Atmospheric Science*, *7*(1), 36. https://doi.org/10.1038/s41612-024-00585-6
Sun, Y., Zhu, S., Wang, D., Duan, J., Lu, H., Yin, H., et al. (2024). Global supply chains amplify economic costs of future extreme heat risk. *Nature*, *627*(8005), 797–804. https://doi.org/10.1038/s41586-024-07147-z
Teuling, A. J., Hirschi, M., Ohmura, A., Wild, M., Reichstein, M., Ciais, P., et al. (2009). A regional perspective on trends in continental evaporation. *Geophysical Research Letters*, *36*(2). https://doi.org/10.1029/2008GL036584
Vanos, J., Guzman-Echavarria, G., Baldwin, J. W., Bongers, C., Ebi, K. L., & Jay, O. (2023). A physiological approach for assessing human survivability and liveability to heat in a changing climate. *Nature Communications*, *14*(1), 7653. https://doi.org/10.1038/s41467-023-43121-5
Vecellio, D. J., Wolf, S. T., Cottle, R. M., & Kenney, W. L. (2022). Utility of the Heat Index in defining the upper limits of thermal balance during light physical activity (PSU HEAT Project). *International Journal of Biometeorology*, *66*(9), 1759–1769. https://doi.org/10.1007/s00484-022-02316-z
Vogel, M. M., Hauser, M., & Seneviratne, S. I. (2020). Projected changes in hot, dry and wet extreme events' clusters in CMIP6 multi-model ensemble. *Environmental Research Letters*, *15*(9), 094021. https://doi.org/10.1088/1748-9326/ab90a7
Vogel, M. M., Orth, R., Cheruy, F., Hagemann, S., Lorenz, R., Hurk, B. J. J. M., & Seneviratne, S. I. (2017). Regional amplification of projected changes in extreme temperatures strongly controlled by soil moisture-temperature feedbacks. *Geophysical Research Letters*, *44*(3), 1511–1519. https://doi.org/10.1002/2016GL071235
Willett, K. M., & Sherwood, S. (2010). Exceedance of heat index thresholds for 15 regions under a warming climate using the wet-bulb globe temperature. *International Journal of Climatology*, *32*(2), 161–177. https://doi.org/10.1002/joc.2257
Yaglou, C. P., & Minard, D. (1957). Control of heat casualties at military training centers. *A.M.A. Archives of Industrial Health*, *16*(4), 302–316.
Zhang, K., Cao, C., Chu, H., Zhao, L., Zhao, J., & Lee, X. (2023). Increased heat risk in wet climate induced by urban humid heat. *Nature*, *617*(7962), 738–742. https://doi.org/10.1038/s41586-023-05911-1
Zhang, Y., Boos, W. R., Held, I., Paciorek, C. J., & Fueglistaler, S. (2024). Forecasting tropical annual maximum wet-bulb temperatures months in advance from the current state of ENSO. *Geophysical Research Letters*, *51*(7), e2023GL106990. https://doi.org/10.1029/2023GL106990
Zhang, Y., & Fueglistaler, S. (2020). How tropical convection couples high moist static energy over land and ocean. *Geophysical Research Letters*, *47*(2). https://doi.org/10.1029/2019GL086387
Zhang, Y., Held, I., & Fueglistaler, S. (2021). Projections of tropical heat stress constrained by atmospheric dynamics. *Nature Geoscience*, *14*(3), 133–137. https://doi.org/10.1038/s41561-021-00695-3
Zhu, J., Wang, S., Zhang, B., & Wang, D. (2021). Adapting to changing labor productivity as a result of intensified heat stress in a changing climate. *GeoHealth*, *5*(4). https://doi.org/10.1029/2020GH000313